Convex reductions in symplectic systolic optimization

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Abstract

The pentagon product H of Haim–Kislev and Ostrover violates Viterbo's volume–capacity inequality. We give an explicit twenty-facet intersection $H_{3/10}$ with exactly computed EHZ capacity and volume, and systolic ratio $1.051914\ldots > (3+\sqrt{5})/5 = \rho(H)$. Yet H maximizes the ratio over its entire affine orbit, with equality only at symplectic similarities. Separately, the same cut family has strictly positive first variation at H; any seven of its ten cuts yield a seventeen-facet ascent. Thus H is not a local maximizer among all convex four-bodies.

For a product of two triangles, every affine position retaining Lagrangian factor planes has ratio at most $2/3$, sharply, while an explicit rational non-Lagrangian position has ratio $18/25$. This does not locate its full affine optimum.

For four-simplices we give a graph proof and equality classification for the sharp bound $c^2_{\mathrm{EHZ}}/(2\,\mathrm{Vol}_4) \le 3/4$, previously established by Haim–Kislev. For six-simplices an exact computer-assisted polyhedral decomposition proves the sharp bound $c^3_{\mathrm{EHZ}}/(6\,\mathrm{Vol}_6) \le (945+75\sqrt{5})/1922$ and a unique equality class. In a different convex reduction, we prove a sharp difference-volume bound for intersections of two confocal paraboloids in every dimension. Its planar case, together with reciprocal-body geometry, gives $\mathrm{area}(\sigma K - \sigma K) \le 5\,\mathrm{area}(\sigma K)$ for complex squaring $\sigma(z) = z^2$ and symmetric convex K; this yields a four-dimensional capacity bound under right quaternionic I-symmetry. Finally, every convex partner of a corner-truncated parallelogram satisfies the strict Viterbo inequality, although an explicit family disproves the proposed optimizer.

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1 Convex reductions after the pentagon counterexample

Let $c = c_{\mathrm{EHZ}}$ be the Ekeland–Hofer–Zehnder capacity, normalized by $c(B^{2n}(r)) = \pi r^2$. In dimension four we write

$$\rho(K) = \frac{c(K)^2}{2\,\mathrm{Vol}_4(K)}.$$

The ball has $\rho = 1$. Haim–Kislev and Ostrover exhibited a convex Lagrangian product of two regular pentagons [7] with $\rho = (3+\sqrt{5})/5 > 1$. Thus the bound $\rho \le 1$ cannot hold for all convex bodies. This changes the optimization question rather than removing it: which classes retain sharp inequalities, what are their extremizers, and which representations make their capacities tractable?

We use the finite EHZ action formula, convex difference bodies, and planar translation covers to study these questions. The reductions keep track of their geometric scope: an affine orbit differs from a Hausdorff neighborhood, and a sharp planar constant need not be sharp after a symplectic transfer.

1.1 Results and their scopes

Section 2 optimizes over every linear image of the pentagon product. Such a map can destroy the Lagrangian factor planes. The action-body representation retains these deformations and supplies the sharp bound and its complete equality class. Section 3 shows, for a different product of two triangles, that leaving Lagrangian position can strictly improve the ratio.

Section 4 studies a different deformation space: simultaneous inward facet splittings. An exact capacity and volume computation for its twenty-facet member $H_{3/10}$ gives ratio $1.051914\ldots$, strictly above the pentagon product. A separate first-variation calculation establishes improving bodies arbitrarily close to the product; its weighted response shows that any seven of the ten cuts suffice for a seventeen-facet ascent. For arbitrary affine cuts constant along their own facet characteristics, one or two added rows have nonpositive first variation; this is not a twelve-facet neighborhood theorem. The appendices prove the general variation formula, a quantitative near-zero action remainder, and the two-cut obstruction.

Section 5 gives a graph proof of the sharp four-simplex bound $\rho \le 3/4$. Haim–Kislev established the bound in her 2019 thesis; the graph proof and equality description are the contributions here. Sections 6–7 prove the sharp global six-simplex bound. Its polyhedral completeness step uses exact cdd and PPL computation, with replay code retained alongside the paper.

Section 8 proves the all-dimensional two-paraboloid difference-volume theorem, then the sharp planar inequality for squared bodies. The capacity consequence uses Chaidez–Tanny's squared Finsler codisks. Neither the sharp planar constant nor its limiting extremizers establish sharpness of the resulting four-dimensional constant $5/4$ for ρ.

Section 9 proves the strict inequality for every convex partner of a corner-truncated parallelogram and obtains an open Hausdorff neighborhood of second factors with the same property. An explicit capacity-one competitor disproves the optimizer proposed in [2]; its ratio remains below one.

1.2 Conventions and the finite capacity formula

We use a fixed standard symplectic form ω_0 on $\mathbb{R}^4$. Interleaved coordinates have $\omega_0 = dz_1 \wedge dz_2 + dz_3 \wedge dz_4$; product coordinates $(q,p) \in \mathbb{R}^2 \times \mathbb{R}^2$ instead have $\omega_0 = \sum_j dq_j \wedge dp_j$. These are the same convention after the indicated coordinate permutation.

For a bounded full-dimensional polytope containing the origin, write $K = \{z : a_i \cdot z \le 1\}$. The Haim–Kislev formula [6] takes the maximum, over facet orders and weights satisfying

$$\beta_i \ge 0, \qquad \sum_i \beta_i = 1, \qquad \sum_i \beta_i a_{\sigma(i)} = 0,$$

of the ordered quadratic action

$$Q = \sum_{i<j} \beta_i \beta_j \omega_0(a_{\sigma(i)}, a_{\sigma(j)}), \qquad c(K) = \frac{1}{2Q_{\max}}.$$

The displayed traversal convention reverses the alternative order used in some statements of the formula. Maximization over all orders makes the scalar value identical. A single feasible word gives an upper bound for capacity. Equality or a lower bound requires completeness of the candidates, which will be proved separately wherever it is used.

2 The pentagon counterexample in its full affine class

The pentagon counterexample is naturally presented as a Lagrangian product. An arbitrary linear image need not retain that feature: its two factor planes can cease to be Lagrangian. We optimize over this larger class. The proof records all affine deformations by a skew form, then projects the resulting six-dimensional capacity constraint to a three-dimensional convex body. A small exact certificate places that projection inside the unit ball.

Let $P \subset \mathbb{R}^2$ be the regular pentagon with vertices $(\cos(2\pi k/5), \sin(2\pi k/5))$, and put

$$H = P \times R(-\pi/2)P \subset \mathbb{R}^2_q \times \mathbb{R}^2_p.$$

Haim–Kislev and Ostrover computed $\rho(H) = (3+\sqrt{5})/5$ [7].

Theorem 2.1 (Affine optimality of the HKO product)**.** *For every $A \in GL(4,\mathbb{R})$,*

$$\rho(AH) \le \rho(H) = \frac{3+\sqrt{5}}{5}.$$

Equality holds if and only if $AH = \lambda SH$ for some $\lambda > 0$ and $S \in \mathrm{Sp}(4,\mathbb{R})$. The same statement holds after allowing translations of the bodies.

The class includes separate affine changes of the two pentagon factors and linear maps mixing their planes. It does not include general perturbations of the ten facets or bodies with a different affine shape.

2.1 Affine deformation through an action body

Use product coordinates and the matrix

$$J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}, \qquad \omega_0(a,b) = a^{\mathsf{T}} J b.$$

Identify a skew matrix Ω with

$$x = (x_{01}, x_{02}, x_{03}, x_{12}, x_{13}, x_{23}) = (\Omega_{01}, \Omega_{02}, \Omega_{03}, \Omega_{12}, \Omega_{13}, \Omega_{23}),$$

and write

$$\mathfrak{p}(x) = -\operatorname{Pf}(\Omega) = x_{02}x_{13} - x_{01}x_{23} - x_{03}x_{12}. \tag{1}$$

In particular, J has coordinates $(0,1,0,0,1,0)$ and $\mathfrak{p}(J) = 1$. The minus sign comes from the product-coordinate order.

For a bounded full-dimensional polytope $K = \{z : a_i \cdot z \le 1\}$, define its *action body* $\mathcal{W}_K \subset \bigwedge^2 \mathbb{R}^4$ as the convex hull of

$$b_{\sigma,\beta} = \sum_{i<j} \beta_i \beta_j \big(a_{\sigma(i)} \wedge a_{\sigma(j)}\big), \quad \beta_i \ge 0, \quad \sum_i \beta_i = 1, \quad \sum_i \beta_i a_{\sigma(i)} = 0, \tag{2}$$

where σ ranges over words of distinct facets. The coordinate of $a \wedge b$ indexed by ij is $a_i b_j - a_j b_i$, so its pairing with x is $a^{\mathsf{T}}\Omega b$.

Lemma 2.2 (Affine capacity formula)**.** *The body $\mathcal{W}_K$ is compact, centrally symmetric, and has nonempty interior. Its support function satisfies*

$$c_{\mathrm{EHZ}}(AK) = \frac{1}{2h_{\mathcal{W}_K}(A^{-1}JA^{-\mathsf{T}})} \qquad (A \in GL(4,\mathbb{R})). \tag{3}$$

For $\det A > 0$ *and* x *the coordinates of* $A^{-1}JA^{-\mathsf{T}}$*,*

$$\rho(AK) = \frac{\mathfrak{p}(x)}{8\,\mathrm{Vol}_4(K)h_{\mathcal{W}_K}(x)^2}. \tag{4}$$

Proof. Each word has a compact feasible weight set, and there are finitely many words. Reversing the word and its weights negates its bivector, proving central symmetry. Under $K \mapsto AK$, the normals become $A^{-\mathsf{T}}a_i$, while the feasible weights remain exactly the same. Their ordered action is the pairing in (2) with $A^{-1}JA^{-\mathsf{T}}$. The Haim–Kislev formula [6] therefore gives (3). Its source convention pairs the later facet with the earlier one; reversing every word and its weights gives the displayed convention. Inserting or deleting zero weights identifies the subset-word formula with the full-permutation formula.

To prove nonempty interior, choose a strictly positive feasible vector on all the normals. Such a vector exists because the normals positively span $\mathbb{R}^4$, so zero lies in the interior of their convex hull. If a skew form annihilates $\mathcal{W}_K$, every ordering of this vector has zero action. Interchanging adjacent facets i, j changes the action by $2\beta_i\beta_j a_i^{\mathsf{T}}\Omega a_j$. All these pairings must vanish. Since the normals span $\mathbb{R}^4$, the form is zero. Thus $\mathcal{W}_K$ spans the bivector space; its symmetry gives nonempty interior. In particular its support function is a norm.

Finally,

$$\mathrm{Pf}(A^{-1}JA^{-\mathsf{T}}) = \frac{\mathrm{Pf}(J)}{\det A} = -\frac{1}{\det A}.$$

Combining this identity with $\mathrm{Vol}_4(AK) = \det A\,\mathrm{Vol}_4(K)$ proves (4). □

The lemma replaces optimization over matrices A by optimization of a quadratic Pfaffian on the polar of a fixed convex body. The capacity constraint retains its full meaning even when the factor planes are no longer Lagrangian. A finite collection of feasible witnesses gives only some supporting inequalities of this polar, which is sufficient for an upper bound if those inequalities have the right geometry.

2.2 A three-dimensional projection inside the unit ball

For the rest of the section write $\mathcal{W} = \mathcal{W}_H$. Set

$$t = \tan(\pi/5), \quad s = \sqrt{5} = \frac{5-t^2}{2}, \quad c_0 = 5t - \frac{t^3}{2}, \quad q_0 = \frac{1}{2c_0}, \quad V_0 = \frac{25(5+s)}{32}.$$

Here $c_0 = c_{\mathrm{EHZ}}(H)$ and $V_0 = \mathrm{Vol}_4(H)$, so

$$\frac{1}{8V_0q_0^2} = \frac{3+s}{5}. \tag{5}$$

Consider the full normalized action polar

$$\mathcal{C} = \{x : h_{\mathcal{W}}(x) \le q_0\}.$$

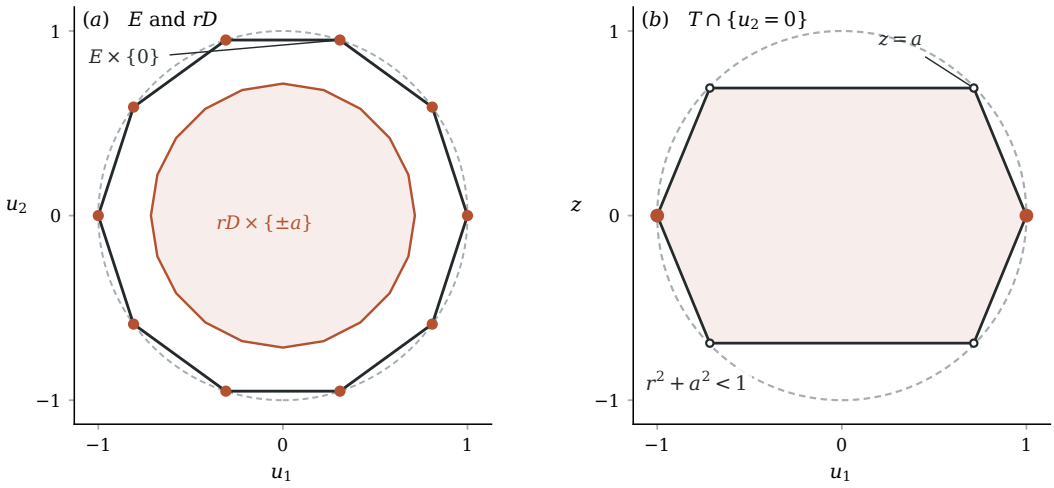


Figure 1: The outer projection body T: its middle and end polygons in plan view, and an axial section inside the unit-ball section. Only the ten equatorial vertices meet the sphere.

Define two linear maps to $\mathbb{R}^3$ by

$$U(x) = \frac{1}{2}(x_{02} + x_{13},\ x_{03} - x_{12},\ x_{01} - x_{23}), \tag{6}$$
$$V(x) = \frac{1}{2}(x_{02} - x_{13},\ x_{03} + x_{12},\ x_{01} + x_{23}).$$

These are coordinates on the two three-dimensional summands of the skew forms. Direct expansion gives

$$\mathfrak{p}(x) = |U(x)|^2 - |V(x)|^2. \tag{7}$$

Consequently a unit-ball bound on $U(\mathcal{C})$ controls the Pfaffian.

The outer body used for that bound is elementary. Let D be the regular 20-gon with vertices at angles $k\pi/10$, of circumradius one, and let $E \subset D$ be the regular decagon consisting of the even-indexed vertices. Put

$$a = \frac{5 - \sqrt{5}}{4}, \qquad r = \frac{143}{200}, \qquad T = \operatorname{conv}\big((E \times \{0\}) \cup (rD \times \{a\}) \cup (rD \times \{-a\})\big). \tag{8}$$

The rational number r supplies convenient slack in the certificate; it is not asserted to be an optimal auxiliary constant.

Proposition 2.3 (Projection certificate)**.** *The full action polar satisfies*

$$U(\mathcal{C}) \subset T \subset \overline{B}^3(1).$$

Moreover $T \cap S^2$ consists of exactly the ten vertices of $E \times \{0\}$.

Proof. First describe T without computation. Since $rD \subset E$, its slice at height $|z| = a\tau$, for $0 \le \tau \le 1$, is $(1-\tau)E + \tau rD$. One way to verify this slice identity is to replace a mixture of the upper and lower copies of rD by a point of E at height zero; the remaining mixture has the stated form. The edge normals of the planar Minkowski sum are the union of the edge normals of its two summands. Thus a complete halfspace description of T is $|z| \le a$, together with

$$n \cdot u + \frac{d_0 - d_1}{a}|z| \le d_0, \qquad d_0 = h_E(n), \quad d_1 = r h_D(n), \tag{9}$$

for every edge-normal direction of E or D. There are 62 halfspaces: two horizontal ones and two for each of 30 planar directions. For exact coordinates, the edge normals of E can be taken to be the odd vertices of D, and those of D to be sums of adjacent vertices of D.

The vertices in the middle level of T have norm one, whereas all the vertices in its upper and lower levels satisfy

$$|(ru, \pm a)|^2 = r^2 + a^2 < 1.$$

Convexity proves $T \subset \overline{B}^3(1)$. Strict convexity of the ball shows that its unit-sphere contacts are exactly the ten middle vertices.

It remains to place $U(\mathcal{C})$ in these halfspaces. The finite certificate described below supplies 18 feasible action bivectors $q_0 b_j \in \mathcal{W}$. Write $Lx = U(x)$. For four representative halfspaces $h_0 + h \cdot y \geq 0$, with $h_0 > 0$, it gives coefficients $\lambda_j \geq 0$ satisfying

$$\sum_j \lambda_j = 1, \qquad \sum_j \lambda_j b_j = -\frac{h^{\mathsf{T}} L}{h_0}. \tag{10}$$

Every $x \in \mathcal{C}$ satisfies $b_j \cdot x \leq 1$. Hence (10) implies the representative halfspace inequality at $U(x)$.

The remaining halfspaces follow by symmetry. The projection $U(\mathcal{C})$ is invariant under the group generated by

$$(u, z) \longmapsto (R(2\pi/5)u, z), \qquad (u_1, u_2, z) \longmapsto (u_1, -u_2, z), \qquad y \longmapsto -y.$$

Indeed, the first map is induced by congruence of Ω with the $-2\pi/5$ rotation of the q pentagon. The second is induced by the negative of congruence with $\mathrm{diag}(1, -1, -1, 1)$, the simultaneous horizontal reflection of the q pentagon and vertical reflection of its partner. The last is $\Omega \mapsto -\Omega$. The two congruences permute the facet normals of H, and therefore permute the feasible words defining $\mathcal{W}$; negation preserves its support function by word reversal. Substitution in (6) gives the claimed maps on U. The four representatives have orbits of sizes $2, 20, 20, 20$, covering all 62 halfspaces. The exact verification also checks these normal permutations, projection identities, and orbit coverage.

The symmetry argument applies to the *full* action polar $\mathcal{C}$. The polyhedron defined only by the selected 18 witness inequalities need not be invariant under this group. No such invariance is used. This proves $U(\mathcal{C}) \subset T$. □

2.3 The finite predicate and its reproduction

All finite data lie in the ordered field $F = \mathbb{Q}(t)$, where $t^4 - 10t^2 + 5 = 0$ and $0 < t < 1$. To specify the geometry used by the certificate, put

$$\alpha = \frac{3 - s}{2}, \qquad \beta = \frac{t(1 + s)}{2}, \qquad \delta = s - 1.$$

Define the five planar normals

$$u_0 = (1, t), \quad u_1 = (-\alpha, \beta), \quad u_2 = (-\delta, 0), \quad u_3 = (-\alpha, -\beta), \quad u_4 = (1, -t).$$

The ten four-dimensional normals are $a_i = (u_i, 0)$ and $a_{5+i} = (0, R(-\pi/2)u_i)$ for $0 \leq i < 5$. For each of the 18 witnesses the verifier checks distinctness of the facet indices, nonnegativity and normalization of the weights, closure, and the exact bivector identity (2). It then checks the four positive relations (10). The words and field coefficients are stored in `certificates/affine-hko/certificate.json`; they are finite proof inputs rather than requests to run an optimizer.

For a compact description of the four relations, let $C = \cos(\pi/10)$, $S = \sin(\pi/10)$ and $\eta = S/(1+C)$. Table 1 gives the representative inequalities, their support sizes, and simple exact lower bounds on every nonzero weight. The first relation averages the two pure-factor five-facet witnesses with weights $1/2, 1/2$. The remaining coefficients are longer elements of F; their complete values are retained in the certificate.

Representative halfspace in (u_1, u_2, z)	Witnesses	Weight lower bound	Orbit size
$a + z \geq 0$	2	$1/2$	2
$C - u_2 + (C - r)z/a \geq 0$	4	$1/40$	20
$1 - u_1 - \eta u_2 + (1 - r)z/a \geq 0$	6	$1/200$	20
$1 - u_1 - \eta u_2 - (1 - r)z/a \geq 0$	6	$1/30$	20

Table 1: Four exact convex combinations imply the complete outer-body containment by symmetry. The bounds in the third column are not substitutes for the exact identities.

The standalone SageMath program verifies the finite data from one JSON input file. It reconstructs the facet normals, all witness bivectors, the outer vertices and halfspaces, and the symmetry actions. It accepts exactly the predicates used above: 18 feasible witnesses, four positive relations, coverage of 62 halfspaces, and the sphere bounds for 50 outer vertices with precisely ten unit contacts. It requires neither numerical optimization nor an external polyhedral hull computation. Run `certificates/affine-hko/verify.py` with a Python interpreter providing `sage.all`; it reads the sibling JSON and records an execution summary. A successful run validates the finite predicates; the slice argument, symmetry implication, and affine capacity reduction are the mathematical steps proved in this section.

2.4 Proof of the sharp bound and equality

Proof of Theorem 2.1. First suppose $\det A > 0$ and let $x = A^{-1} J A^{-\mathsf{T}}$ in skew-form coordinates. Put $q = h_{\mathcal{W}}(x)/q_0 > 0$. By definition $x/q \in \mathcal{C}$, so Proposition 2.3 and (7) give

$$\mathfrak{p}(x) = |U(x)|^2 - |V(x)|^2 \leq |U(x)|^2 \leq q^2.$$

Equations (4) and (5) prove the claimed upper bound.

If equality holds, both inequalities are equalities. Thus $V(x) = 0$, and $U(x)/q$ is one of the ten unit-sphere contacts of T. Consequently x/q has zero same-factor entries and cross block

$$\begin{pmatrix} \cos(k\pi/5) & \sin(k\pi/5) \\ -\sin(k\pi/5) & \cos(k\pi/5) \end{pmatrix}, \qquad k = 0, \ldots, 9. \tag{11}$$

Each is $T_0^{-1} J T_0^{-\mathsf{T}}$ for a linear automorphism T_0 of H. Rotations by $2\pi/5$ in one factor give five of the forms; the simultaneous factor reflections used above preserve H and send J to $-J$, giving the other five. Therefore

$$A^{-1} J A^{-\mathsf{T}} = q T_0^{-1} J T_0^{-\mathsf{T}}.$$

The matrix $S_0 = \sqrt{q}\, A T_0^{-1}$ satisfies $S_0 J S_0^{\mathsf{T}} = J$ and is symplectic. Hence $AH = q^{-1/2} S_0 H$. Conversely, symplectic images and positive dilations preserve ρ.

Finally, if $\det A < 0$, compose A on the right with $\operatorname{diag}(1, -1, 1, 1)$. This is a determinant -1 automorphism of H, so it changes the determinant sign without changing AH. The positive-determinant argument applies. Translation invariance gives the last assertion. □

3 A product whose optimal affine position is not Lagrangian

The affine optimality of the pentagon product does not follow from a general principle reducing products to Lagrangian position. Even a product of two triangles has an affine image whose systolic ratio exceeds that of every Lagrangian position of the same product. Here "Lagrangian position" means that both original factor planes become Lagrangian; all independent affine changes within those planes are allowed.

Theorem 3.1. *Let*

$$\Delta = \{(s,t) \in \mathbb{R}^2 : s \le 1,\ t \le 1,\ -s-t \le 1\}, \qquad K = \Delta \times \Delta.$$

Among invertible linear maps B for which both factor planes of BK are Lagrangian, the maximum systolic ratio is $2/3$. The matrix

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1/4 & -1/2 & 0 & -1/4 \end{pmatrix}$$

satisfies

$$c_{\mathrm{EHZ}}(AK) = \frac{27}{10}, \qquad \mathrm{Vol}_4(AK) = \frac{81}{16}, \qquad \rho(AK) = \frac{18}{25} > \frac{2}{3}.$$

Consequently, no maximum of the systolic ratio on the full affine orbit of K occurs in Lagrangian position. By affine equivalence, the same conclusion holds for the product of any two nondegenerate triangles.

The comparison covers *every* Lagrangian position. Both ratios are below one, so the result gives no Viterbo counterexample. It does not compute the full affine maximum.

3.1 Closure and the explicit non-Lagrangian image

Use the conventions of Lemma 2.2. The area of Δ is $9/2$, so $\mathrm{Vol}_4(K) = 81/4$. Its six support-normalized normals are

$$a_0 = e_1, \quad a_1 = e_2, \quad a_2 = -e_1 - e_2, \qquad a_3 = e_3, \quad a_4 = e_4, \quad a_5 = -e_3 - e_4.$$

Closure forces every feasible weight vector to have the form

$$\beta_0 = \beta_1 = \beta_2 = \frac{t}{3}, \qquad \beta_3 = \beta_4 = \beta_5 = \frac{1-t}{3}, \qquad 0 \le t \le 1. \tag{12}$$

Cyclic shifts of an order do not change its action: the change caused by moving its first weighted normal to the end is its pairing with twice the sum of the remaining weighted normals, which vanishes by closure. Thus the 120 orders starting with 0 cover all orders. The endpoints in (12) include smaller supports.

For $x = (1, 1, -2, 1, 1, -1)$, direct multiplication gives

$$\Omega = A^{-1} J A^{-\mathsf{T}}, \qquad \det A = \frac{1}{4}, \qquad \mathfrak{p}(x) = 4.$$

For each order its action is

$$Q_\sigma(t) = \frac{1}{9}\big[at^2 + b(1-t)^2 + dt(1-t)\big], \qquad a, b \in \{-1, 1\}.$$

The following finite table gives the greatest mixed coefficient d in each of the four groups. Each group contains 30 cyclic orders. A witness in every row also makes the bound sharp in that group.

a	b	$\max d$	Attaining order	$\max_{\sigma,t} Q_\sigma(t)$
1	1	4	$(0,1,3,2,5,4)$	$1/6$
1	-1	6	$(0,5,1,3,2,4)$	$5/27$
-1	1	6	$(0,3,2,5,1,4)$	$5/27$
-1	-1	4	$(0,2,5,1,3,4)$	$1/18$

All entries follow by summing the 15 pairings of the listed six normals; the accompanying exact verifier enumerates precisely these 120 orders. Since $t(1-t) \geq 0$, the maximum mixed coefficient bounds the whole group. The four elementary quadratic maxima occur at $t = 1/2, 2/3, 1/3, 1/2$, respectively. Therefore the full Haim–Kislev maximum is

$$h_{\mathcal{W}_K}(x) = \frac{5}{27}.$$

The affine capacity formula gives $c_{\mathrm{EHZ}}(AK) = 27/10$, and the volume and ratio in the theorem follow. The table proves the required global capacity calculation, rather than merely exhibiting one feasible action.

3.2 The entire Lagrangian subclass

For a nondegenerate skew matrix, both diagonal 2×2 blocks vanish if and only if both diagonal blocks of its inverse vanish. Since $B^{\mathsf{T}} J B = -\Omega^{-1}$, the two factor planes are Lagrangian exactly when

$$\Omega = \begin{pmatrix} 0 & M \\ -M^{\mathsf{T}} & 0 \end{pmatrix}, \qquad M \in GL(2,\mathbb{R}).$$

Conversely, every such M occurs, by taking $B = \mathrm{diag}(M^{-1}, I_2)$. Write $h(M)$ for the action support at this cross form. Homogeneity and $|\det B|^{-1} = |\det M|$ give

$$\rho(BK) = \frac{|\det M|}{162 h(M)^2}. \tag{13}$$

The same-factor terms now vanish, so each order has action $t(1-t)$ times a linear form in M. Reversal of an order negates this linear form. Consequently the global maximum occurs at $t = 1/2$, and $h(M)$ is a finite support function. Set

$$C = \{M : h(M) \leq 1/12\}, \qquad u_1 = (1,0)^{\mathsf{T}}, \quad u_2 = (0,1)^{\mathsf{T}}, \quad u_3 = (1,-1)^{\mathsf{T}}.$$

The second finite identity used in the proof is

$$C = \mathrm{conv}\left\{ \pm\frac{3}{2} u_i u_j^{\mathsf{T}} : 1 \leq i,j \leq 3 \right\}. \tag{14}$$

To verify it exactly, form the 120 action rows at weights $1/6$, multiply them by 12, and impose their upper bounds 1. Rational polyhedral reduction gives a bounded four-dimensional polytope with 30 facets and exactly the 18 vertices displayed in (14). The verifier asserts equality of these two vertex sets, not an approximate match. This finite hull identity is the only polyhedral calculation used in the following determinant bound.

Let $D = \mathrm{conv}\{\pm u_1, \pm u_2, \pm u_3\}$. Grouping any convex combination in (14) by its first index writes

$$M = \frac{3}{2} \sum_{i=1}^{3} u_i v_i^{\mathsf{T}}, \qquad v_i \in \alpha_i D, \quad \alpha_i \geq 0, \quad \sum_i \alpha_i = 1.$$

Each determinant of two vertices of D has absolute value at most one. By bilinearity and convexity this implies $|\det(v_i, v_j)| \le \alpha_i\alpha_j$. The two-dimensional Cauchy–Binet identity now yields

$$|\det M| \le \frac{9}{4}\sum_{i<j} |\det(u_i,u_j)|\,|\det(v_i,v_j)| \le \frac{9}{4}\sum_{i<j}\alpha_i\alpha_j \le \frac{3}{4}.$$

The last step follows from $\sum_i \alpha_i^2 \ge 1/3$. Equality is attained by $\alpha_i = 1/3$ and $v_i = u_i/3$, giving

$$M_* = \frac{1}{2}\sum_i u_i u_i^{\mathsf{T}} = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}, \qquad \det M_* = \frac{3}{4}.$$

It also has $h(M_*) = 1/12$: a smaller value would allow dilation within C and contradict the determinant bound. For any nonzero M, the matrix $M/(12h(M))$ belongs to C. Equation (13) therefore gives

$$\rho(BK) \le \frac{144}{162}\cdot\frac{3}{4} = \frac{2}{3},$$

with equality for M_*. This proves the sharp global comparison.

Finally, a full affine maximum exists. By Lemma 2.2, the action support is a norm. The continuous function $|\operatorname{Pf}|$ attains a positive maximum on its compact unit ball. Any maximizing form is nondegenerate and hence is congruent to the standard symplectic form; it is realized by an invertible linear image. Since its ratio is at least $18/25$, no maximizing position has Lagrangian factor planes.

Exact checks and scope. The retained verifier `certificates/triangle-position/exact_triangles_compact.py` uses rational arithmetic and Sage's PPL backend to reconstruct both finite identities from the six normals. It verifies the four action-coefficient groups, the explicit matrix, and the 18-vertex identity. An independent calculation also checks the global determinant maximum by enumerating stationary points on faces of dimensions at most two. The proof above replaces that larger face calculation by Cauchy–Binet. These checks do not establish priority in the literature. Published bounds for triangle–convex Lagrangian products concern a different comparison class. The bounded source comparison is recorded at `reviews/triangle-position-prior-comparison.md`; it does not establish priority.

4 A simultaneous-cut improvement of the pentagon product

The explicit twenty-facet body $H_{3/10}$ in Theorem 4.4 has ratio $1.051914\ldots > (3+\sqrt{5})/5$, although the pentagon product maximizes its entire affine orbit. The finite-parameter capacity computation is separate from local behavior at the product. We prove that the product is not a local maximum when the number of facets is unrestricted by studying the right derivative of a fixed twenty-row family. The calculation combines the simultaneous-splitting derivative with a reduction tailored to its cut profiles and an exhaustive exact calculation of the remaining shortest-word responses.

The complex structure $\mathbb{J}$ below is the one in the appendix, whose matrix is the negative of the skew-form matrix J in Section 2.1.

Let $P \subset \mathbb{R}^2$ be the regular pentagon with vertices $(\cos(2\pi k/5), \sin(2\pi k/5))$, $0 \le k < 5$, and let $J_2(x,y) = (-y,x)$. In coordinates (q,p) put

$$H = P \times_L (-J_2 P), \qquad \mathbb{J}(q,p) = (-p,q), \qquad \tau = \tan(\pi/5).$$

Let n_i be the support-normalized outward normal of the edge from vertex i to vertex $i+1$ of P, with indices taken modulo five, and set $m_i = -J_2 n_i$. Thus $n_0 = (1, \tau)$ and $|n_i| = |m_i| = \sqrt{5} - 1$. The ten normals of H are $a_i = (n_i, 0)$ and $a_{5+i} = (0, m_i)$. Define

$$D_i = \left(-\frac{3}{5} n_i, \frac{J_2 n_i}{|n_i|} \right), \qquad D_{5+i} = \left(-\frac{J_2 m_i}{|m_i|}, -\frac{3}{5} m_i \right), \tag{15}$$

$$H_t = \{ x : a_j \cdot x \le 1,\ (a_j + t D_j) \cdot x \le 1,\ 0 \le j < 10 \}. \tag{16}$$

Every old inequality is retained, so $H_t \subset H$ for $t \ge 0$. For sufficiently small t these are full-dimensional bounded polytopes and $H_t \to H$ in Hausdorff distance. Equivalently,

$$H_t = H \cap L_t^{-1} H, \qquad L_t = \begin{pmatrix} a I_2 & -b J_2 \\ b J_2 & a I_2 \end{pmatrix}, \quad a = 1 - \frac{3}{5} t, \quad b = \cos(\pi/5) t.$$

There are exactly twenty genuine facets for all sufficiently small $t > 0$. Indeed, on the relative interior of each old facet, $D_i \cdot x$ takes both positive and negative values on open patches. A negative patch retains the old facet with every competing inequality strict. At a positive patch, the point $x/(1 + t D_i \cdot x)$ lies on the new hyperplane, strictly satisfies the old inequality, and keeps all other inequalities strict for small t. Thus each old and each new row supports a facet. Write $\mathrm{sys}(K) = c_{\mathrm{EHZ}}(K)^2 / (2 \operatorname{Vol}_4(K))$.

Theorem 4.1 (Strict inward ascent)**.** *For the family* (16)*,*

$$\left. \frac{d}{dt} \right|_{0+} \log \mathrm{sys}(H_t) = \frac{20 + 64\sqrt{5}}{9375} > 0. \tag{17}$$

Consequently $\mathrm{sys}(H_t) > \mathrm{sys}(H)$ *for every sufficiently small positive* t*. In particular,* H *is not a local maximum among convex bodies in the Hausdorff topology.*

The statement concerns twenty facets and does not contradict the separately proved local maximum among convex four-polytopes with at most eleven genuine facets.[1] The existence of a small improving interval follows from the positive derivative. Proposition B.1 in the quantitative appendix also supplies an explicit near-zero subinterval for this family.

4.1 The cut profile and the repeated-word issue

On each original facet the inward profile is

$$f_i = (\ell_i - 3/5)_+, \qquad \ell_i(q,p) = \frac{J_2 n_i}{|n_i|} \cdot p, \qquad f_{5+i} = (\ell_{5+i} - 3/5)_+, \quad \ell_{5+i}(q,p) = -\frac{J_2 m_i}{|m_i|} \cdot q. \tag{18}$$

These profiles are convex on their respective facets and constant along their own characteristic directions $\mathbb{J} a_i$. The constancy is essential: general convex profiles need not admit the reduction below.

Use the active-word action $Q(\sigma, \beta) = \sum_{r<s} \beta_r \beta_s \langle \mathbb{J} a_{\sigma_r}, a_{\sigma_s} \rangle$, with $\sum \beta_r a_{\sigma_r} = 0$, $\sum \beta_r = 1$, and $\beta_r \ge 0$. Write $Q_* = 1/(2 c_{\mathrm{EHZ}}(H)) = \tau/5$. For a maximizing word, μ_γ denotes its normalized-time occupation measure on the disjoint union of labeled facets. The general derivative theorem uses all finite maximizing words, including repeated labels; an existence theorem for a simple shortest word alone would not suffice.

[1] Computer-assisted proof and independent acceptance record: `evidence/historical-comparisons/f11/README.md`.

Lemma 4.2 (Removal of repeated visits)**.** *Let K be a Lagrangian product of planar polygons with no antipodal facet normals. Suppose each continuous convex facet profile is constant on lines parallel to its own characteristic direction. Every finite maximizing word has a simple maximizing word with at least as large an occupation response. The same supremum therefore bounds the weak closure of the finite-word occupation measures.*

Proof. Delete zero weights. Suppose a full normal $a = (n, 0)$ occurs twice, at positions $r < s$. Transfer weight by $\beta_r(z) = \beta_r + z$, $\beta_s(z) = \beta_s - z$ on $[-\beta_r, \beta_s]$, fixing all other weights. Closure and normalization are preserved. The action is affine in z, since its quadratic coefficient pairs a with itself. The repeated-word polar-loop bound $Q \le Q_*$, together with the interior maximum at $z = 0$, makes this affine function constant. Equality of the normals alone would not imply constancy.

At every interior parameter, the active normals span both factor planes: a positive closed combination without antipodal directions contains at least three distinct directions. The augmented normal matrix has rank five, because averaging a relation $u \cdot a_j + c = 0$ against the positive weights first gives $c = 0$, then $u = 0$. Thus the KKT multiplier $\lambda(z)$ is unique and affine in z. With $T = (2Q_*)^{-1}$, the reconstructed vertices are

$$x_k(z) = -T\lambda(z) + 2T\mathbb{J}\sum_{j<k} \beta_j(z) a_{\sigma_j}.$$

They lie in K on their labeled facets by the reconstruction argument in Theorem A.1.

Choose the cyclic start just before the first duplicate and put $v = 2T\mathbb{J}a$. Every fixed-weight segment translates with derivative u or $u + v$, according to which interval between the duplicates contains it. All q-facet equations give $n_j \cdot u_q = 0$, so $u_q = 0$. The p-facet equations give $m_j \cdot u_p = 0$ outside the interval and $m_j \cdot (u_p + v_p) = 0$ inside. There are at least three distinct p-normal directions; one interval contains two nonparallel ones. Hence $u_p = 0$ or $u_p = -v_p$, and therefore $u = 0$ or $u = -v$.

Both duplicate segments consequently move and change length only along their own characteristic direction. Their profile values c_r, c_s stay constant, and their combined response $(\beta_r + z)c_r + (\beta_s - z)c_s$ is affine. Every other weight is fixed and its segment translates affinely; integrating its convex profile gives a convex function of z. The total response is convex. By continuity it extends to the endpoints, where the zero-weight occurrence is deleted. One endpoint has at least the original response. The p-normal case is identical with the factors exchanged. Iteration removes all repetitions. Continuity on the compact disjoint facet union passes the bound to weak limits of the occupation measures. □

4.2 Complete calculation of the simple-word response

Proposition 4.3 (Exact finite response certificate)**.** *For the profiles* (18)*,*

$$\max_{\text{simple maximizing } \gamma} \int f \, d\mu_\gamma = \frac{1}{125}. \tag{19}$$

The complete maximizing weight family is covered by 190 charts: 50 points on six-label words and 140 closed intervals on seven-label words.

Proof. We describe both the exhaustive reduction and the exact predicates checked by the accompanying portable certificate. The normalized closure polytope in either factor is

$$C_q = \{u \ge 0 : \sum u_i = 1,\ \sum u_i n_i = 0\}, \qquad C_p = \{v \ge 0 : \sum v_i = 1,\ \sum v_i m_i = 0\}.$$

Its vertices have support at most three by rank of the three affine constraints. No support of size one or two closes. Solving all ten three-element supports gives precisely five positive vertices in each factor, with exact weights in $\mathbb{Q}(\tau)$.

For a fixed word, split its weights as $(tu, (1-t)v)$. Same-factor pairings vanish, so $Q = t(1-t)B_\sigma(u,v)$ with B_σ bilinear. Every positive global maximizer has $t = 1/2$. Decomposing u, v into closure vertices expresses its value as a convex combination of triangle-pair actions. Each pair action is at most Q_*, so every pair with positive product coefficient must attain Q_*. Conversely, mixtures supported on a complete bipartite set of winning pairs attain Q_*.

There are $25 \cdot 5! = 3000$ triangle-pair cyclic six-orders. Exact evaluation checks that all have action at most $\tau/5$, with precisely 50 equality words. Every simple support word extends by zero weights to a cyclic ten-order. Fixing label zero first gives all $9! = 362880$ such orders. For each, form the 5×5 graph of its winning induced triangle pairs. Every maximizing weight pair is covered by a maximal complete bipartite subgraph. These are enumerated as follows: for every nonempty row subset R, form its common-neighbor set C, retain it when $C \neq \varnothing$ and R is exactly the common-neighbor set of C. This is precisely the maximality condition. Delete labels absent from the union of its triangle supports and deduplicate by the remaining cyclic word and the two vertex subsets. This loses no maximizing weight or response.

The exhaustive scan gives 79494 orders with winning pairs, 121 distinct nonempty pair graphs, and 190 distinct charts. Fifty charts use one closure vertex in each factor. Seventy use two q-vertices and one p-vertex; seventy use one q-vertex and two p-vertices. No other chart occurs. Thus every nonconstant weight family is an interval $\beta(z) = b + zd$, $0 \le z \le 1$. The catalogue argument needs only the convex hull of the two vertices, not an additional assertion about adjacency in the closure pentagon.

On each chart, solve the rank-five KKT equations exactly. The multiplier and orbit vertices are affine in z, as are the radial coordinates $\ell_i(z)$. Each radial coordinate is constant along its labeled segment. The response therefore has the form

$$R(z) = \sum_i (b_i + zd_i)\max(0, h_i + zk_i).$$

Partition $[0,1]$ at every exact hinge zero. On each resulting interval R is quadratic. Its maximum is attained at an endpoint or at the interior stationary point of a strictly concave quadratic. The checker evaluates all these candidates with exact field arithmetic and isolated signs for the root $0 < \tau < 1$ of $\tau^4 - 10\tau^2 + 5 = 0$. Of the 190 chart maxima, 110 equal zero and 80 equal $1/125$. For example, the chart with word $(0,6,5,2,1,8,4)$ attains $1/125$ at the parameter $z = 3/10 - \tau^2/50$ in the certificate's endpoint ordering. This proves (19). □

The portable replay at `certificates/hko-ascent/README.md` contains the literal normal construction, all positive triangle weights, the complete cyclic-order enumeration, and the exact piecewise-quadratic calculation. It requires Python's standard library and a C++17 compiler. Its checker recomputes all outputs in a fresh output directory and compares them with the retained certificate. A separate Sage-field and Python enumeration reproduced the complete 190-chart set; an independent replay also reproduced all exact response maxima. These checks support the stated finite predicates; the coverage implication is the argument above.

4.3 Volume and the strict derivative

Proof of Theorem 4.1. Each labeled facet has normalized cone-volume mass $1/10$. Its profile is independent of the edge coordinate and cuts a cap from the other pentagon. The cap depth is $\delta = 2/5 < 1 - \cos(2\pi/5)$, so the cap is a triangle with area $\delta^2 \cot(\pi/5)$. The hinge's mean on this triangle is $\delta/3$, and the pentagon area is $A = 5\tau/(1+\tau^2)$. All ten facet

averages agree. Consequently

$$\int f \, d\nu = \frac{\delta^3}{3\tau A} = \frac{8}{625} - \frac{8\tau^2}{9375}. \tag{20}$$

Lemma 4.2 and Proposition 4.3 identify the supremum over all finite maximizing words, and hence their weak closure, with $1/125$. The full-rank support hypothesis of Theorem A.1 holds because both planar normal sets have no antipodal pairs. That theorem applies directly to the fixed twenty-row bank:

$$D^+ \log \mathrm{sys}(H_t) = 4\left(\frac{8}{625} - \frac{8\tau^2}{9375} - \frac{1}{125}\right) = \frac{20 + 64\sqrt{5}}{9375} > 0.$$

Since $H_t \to H$, the definition of a positive right derivative gives the claimed strict improvements arbitrarily close to H. □

4.4 An explicit finite improvement

The same fixed family also gives a visible improvement at a moderate parameter; this does not follow from the positive derivative alone. For H_t in (16), we certify the capacity at $t = 3/10$. The capacity decreases slightly, but the volume decreases by enough to raise the systolic ratio.

Theorem 4.4 (A twenty-facet improvement of HKO)**.** *Let $K = H_{3/10}$ and $\tau = \tan(\pi/5)$. Then K is a simple convex four-polytope with twenty facets and seventy-five vertices, and*

$$c_{\mathrm{EHZ}}(K) = \frac{1019}{410\tau} = \frac{1019}{1025} c_{\mathrm{EHZ}}(H), \qquad \mathrm{Vol}_4(K) = \frac{502711205 + 99664289\sqrt{5}}{130445600}.$$

Its systolic ratio satisfies

$$\rho(K) = 1.0519141388294875\ldots > \rho(H) = \frac{3+\sqrt{5}}{5}.$$

More precisely,

$$\rho(K) - \rho(H) = \frac{7965280 + 4064064\sqrt{5}}{5(502711205 + 99664289\sqrt{5})} > 0.$$

This is a complete capacity computation for one body, rather than an upper bound obtained from one closed characteristic. It does not optimize t or assert a sharp upper bound for all convex bodies.

Proof. Write the twenty support-normalized normals as

$$N = (a_0, \ldots, a_9, a_0 + \tfrac{3}{10}D_0, \ldots, a_9 + \tfrac{3}{10}D_9).$$

Denote the row with label $i \in \{0, \ldots, 19\}$ by n_i. The exact calculation uses the number field $\mathbb{Q}(\tau)$ with $\tau^4 - 10\tau^2 + 5 = 0$ and $0 < \tau < 1$. We describe both the finite predicates and the argument that makes them complete.

Geometry. Solving every invertible four-row subsystem of $Nx \le \mathbf{1}$ and testing all twenty inequalities gives seventy-five vertices. Exactly four linearly independent rows are active at every vertex, and all twenty labels occur. This proves simplicity and that every listed inequality defines a genuine facet. It also determines all nonempty pairwise facet intersections.

Which words need checking. Haim–Kislev's geometric simple-minimizer theorem [6, Theorem 1.5] provides a minimizing characteristic using each facet velocity at most once. Consecutive facet labels i, j must meet and satisfy $\omega(n_i, n_j) \geq 0$: the arriving segment reaches the jth support hyperplane from inside the body. The exact transition graph has 120 directed edges, including zero-pairing edges. Thus some capacity-maximizing word is a simple cycle of this graph.

For a word, let $C\beta = e$ be closure and normalization, and let $Q(\beta) = \frac{1}{2}\beta^{\mathsf{T}} G\beta$. If $z \in \ker C$ has $z^{\mathsf{T}} G z > 0$, the word cannot have an interior maximum. Zero extension of z gives the same obstruction for every larger word containing it in cyclic order. The certificate verifies 77 927 symmetry representatives of such obstructions by exact nullspace and quadratic-form calculations. Cyclic invariance on the constraint kernel follows from closure; the twenty symmetry actions are checked exactly.

An integer depth-first traversal then enumerates all simple cycles, stopping only at a certified obstructed prefix. Every remaining cycle occurs in the checked list. The traversal covers 43 110 cycles, represented by 2326 words after symmetry, with no unaccounted cycle. This coverage check uses no floating-point curvature decision.

The scalar maximum on each remaining word. Deleting zero weights from the geometric maximizing word places it in the relative interior of its searched active cycle. Thus boundary maxima are covered by their positive support. After removing dependent equality constraints, every interior maximum obeys

$$G\beta + C^{\mathsf{T}}\lambda = 0, \qquad C\beta = e.$$

For an invertible KKT matrix the solution is computed exactly; a negative weight excludes that stationary point. Singular systems are handled as affine families. Their objective is constant: a kernel element (z, η) satisfies $Cz = 0$, $Gz + C^{\mathsf{T}}\eta = 0$, and hence $e \cdot \eta = 0$. Consequently $Q = -e \cdot \lambda/2$ has the same value throughout the family.

Of the 1030 singular families, 984 have objective at most the target $205\tau/1019$. Another 21 are infeasible by the elementary inclusion

$$\frac{25}{28}H \subset K, \qquad Q_{\max}(K) \leq \frac{\tau}{5}\left(\frac{28}{25}\right)^2.$$

Indeed, every moving normal has support at most $1 + (2/5)(3/10) = 28/25$ on H. The remaining 25 families are proved infeasible by exact vertex enumeration in their free parameters. This is exhaustive: their nonnegative weights would lie in the compact simplex, the parameter map is injective, and any nonempty bounded parameter polyhedron has a vertex determined by independent active zero-weight equations. Thus no singular or flat family is omitted. The resulting global upper bound is

$$Q_{\max}(K) \leq \frac{205}{1019}\tau.$$

Equality is witnessed by the word $(0, 1, 6, 7, 3, 12, 9)$ with positive weights

$$\frac{1}{8152}(1271{+}123\tau^2,\ 697{+}205\tau^2,\ 965{+}205\tau^2,\ 1070{+}190\tau^2,\ 1804{-}328\tau^2,\ 400,\ 1945{-}395\tau^2).$$

The exact checks verify closure, normalization, positivity, and objective $205\tau/1019$. The Haim–Kislev formula therefore gives the claimed capacity.

Volume and ratio. For a simple four-polytope, Lawrence's formula [8] is

$$\mathrm{Vol}_4(K) = \frac{1}{4!}\sum_v \frac{(\ell \cdot v)^4}{|\det N_{I(v)}| \prod_{j=1}^4 (\ell N_{I(v)}^{-1})_j},$$

where $I(v)$ lists the four active normals and ℓ is nonconstant on every edge. Exact evaluations with $\ell = (1, 2, 4, 8)$ and $(1, 3, 9, 27)$ have nonzero denominators and agree with the stated volume. Substituting the capacity and simplifying in $\mathbb{Q}(\sqrt{5})$ proves the displayed positive ratio difference. □

The portable certificate at `certificates/hko-t03/README.md` reconstructs the twenty normals, validates the integer transition graph, checks every curvature exclusion and stationary family, verifies the positive attaining weights, and recomputes the volume. The coverage proof is independent of the numerical search that proposed the word list. The computation concerns the exact algebraic body $H_{3/10}$; it does not replace it by rounded input data.

4.5 Seventeen facets and the exact response to unequal cut speeds

The twenty-row family admits a sharper facet-count conclusion. Keep the ten profiles of (18) fixed, but give their new rows nonnegative speeds $w = (w_0, \dots, w_9)$:

$$H_{t,w} = \{x : a_i \cdot x \le 1,\ (a_i + t w_i D_i) \cdot x \le 1,\ 0 \le i < 10\}.$$

Its profile on facet i is $w_i f_i$. Write

$$\kappa = \int_{F_i} f_i \, d\nu = \frac{8(5+\sqrt{5})}{46875};$$

this value is independent of the facet and includes its cone-volume mass $1/10$.

Proposition 4.5 (Exact weighted response)**.** *For every $w_i \ge 0$, the supremum over all finite maximizing words and their weak occupation limits is*

$$\sup_{\gamma} \sum_{i=0}^{9} w_i \int_{F_i} f_i \, d\mu_\gamma = \frac{\max_i w_i}{125}. \tag{21}$$

Consequently

$$D^+ \log \mathrm{sys}(H_{t,w}) = 4 \left(\kappa \sum_i w_i - \frac{\max_i w_i}{125} \right). \tag{22}$$

Proof. Nonnegativity and Proposition 4.3, together with Lemma 4.2, give the upper bound

$$\sum_i w_i \int_{F_i} f_i \, d\mu_\gamma \le (\max_i w_i) \int f \, d\mu_\gamma \le \frac{\max_i w_i}{125}.$$

For the reverse bound we exhibit a maximizing word whose entire response is concentrated on one facet. Put $u = \tau^2$ and take

$$\sigma = (0, 6, 5, 2, 1, 8, 4),$$
$$\beta = \frac{1}{200}(17 + 5u,\ 25 + 5u,\ 25 + 5u,\ 44 - 8u,\ 8,\ 50 - 10u,\ 31 + 3u).$$

The weights are strictly positive, sum to one, close the normal word, and give $Q = \tau/5$. Its KKT reconstruction has constant radial hinge arguments, in this order,

$$\ell_{\sigma_r} - \frac{3}{5} = \frac{1}{40}(-59 + 5u,\ -9 - 5u,\ -9 - 5u,\ -9 - 5u,\ 8,\ -9 - 5u,\ -45 + 3u).$$

Since $0 < u < 1$, only the facet labeled 1 contributes. Its weight is $1/25$ and its hinge value is $1/5$, giving response $1/125$ on that facet and zero on all others. These are literal exact

predicates checked by the supplementary witness replay; the same point is chart 11 of the complete catalogue, at parameter $3/10 - \tau^2/50$.

Simultaneous rotation of the q- and p-planes by $2\pi/5$ preserves H, the symplectic form, and this labeled profile family. The symplectic involution

$$S(q,p) = (J_2 p, -J_2 q)$$

also preserves H and exchanges facet i with facet $5+i$, including their cut velocities. These symmetries act transitively on the ten facets. Thus every facet admits a shortest word with response $1/125$ concentrated there. Choose a facet with maximal w_i to obtain the lower bound in (21). The derivative formula follows from Theorem A.1 and the volume response $\kappa \sum_i w_i$. □

Corollary 4.6 (A seventeen-facet ascent)**.** *Choose any seven of the ten cuts in (15), with unit speed, and omit the other three new rows. For all sufficiently small positive t the resulting body has seventeen genuine facets, and*

$$D^+ \log \operatorname{sys}(H_{t,w}) = \frac{4(56\sqrt{5} - 95)}{46875} > 0.$$

Within the cone of nonnegative speeds on these fixed ten profiles, seven new rows is the smallest number that can give a positive first variation.

Proof. The exact comparisons are

$$6\kappa = \frac{240 + 48\sqrt{5}}{46875} < \frac{375}{46875} = \frac{1}{125} < \frac{280 + 56\sqrt{5}}{46875} = 7\kappa.$$

Seven unit speeds therefore give the displayed positive derivative. Every selected new row and every old row is a genuine facet by the open-patch argument following (16), so the facet count is seventeen. If instead at most six speeds are nonzero and $w \neq 0$, then $\sum_i w_i \leq 6 \max_i w_i$, making (22) strictly negative. This proves the claimed minimal count for this fixed profile cone; it does not exclude ascent using different profiles or a different perturbation with fewer new facets. □

4.6 Two characteristic-invariant cuts

The seven-cut threshold above concerns ten fixed radial profiles. A different obstruction applies to any two affine hinges that are constant along their own facet characteristic directions. Affine offsets, dependence on the facet-edge coordinate, and unequal cut speeds are allowed. For a nonnegative function f on the disjoint labeled facet union, put

$$R(f) = \sup_{\gamma} \int f \, d\mu_\gamma,$$

where the supremum runs over all finite maximizing words, equivalently their weak occupation limits.

Proposition 4.7 (Two invariant hinges)**.** *On one HKO facet F_i, let $f_i(x) = (D \cdot x)_+$, with f zero on the other labeled facets and $D \cdot \mathbb{J} a_i = 0$. Then*

$$R(f) \geq 2\nu(f). \tag{23}$$

Consequently, retain all ten original inequalities of H and add at most two rows $(a_i + tD) \cdot x \leq 1$ satisfying this characteristic invariance. Their one-sided first logarithmic systolic variation at $t = 0$ is nonpositive, whether the rows concern the same facet or different facets.

Appendix C proves the one-hinge bound from four literal shortest-characteristic families and a scalar hinge inequality, then handles both placements of two cuts. This is a first-order obstruction for invariant affine hinges. It does not prove local maximality among twelve-facet bodies, decide zero-derivative directions at higher order, or cover non-invariant hinges.

Quantitative near-zero control. A uniform quadratic remainder for the true global optimal action makes the positive first variation effective. For the original family with all ten unit cuts it certifies $\rho(H_t)/\rho(H) > 1 + t/200$ when $0 < t \le 10^{-30}$. The pivot-and-rounding proof in Appendix B illustrates a quantitative sensitivity method; this bound is not asserted for the seventeen-facet subfamilies.

5 The simplex bound through a triangle-free graph

Haim–Kislev proved the sharp bound $c_{\mathrm{EHZ}}(K)^2/(2\operatorname{vol}_4 K) \le 3/4$ for every four-dimensional simplex in her 2019 MSc thesis [5, Theorem B and Section 4]. Her proof uses pairing coordinates and a substantial sign-case analysis. Later computations of Chaidez–Hutchings [3, Section 1.5] give equality examples, while Leipold [9, Section 12.4] reports numerical simplex values. We give a different proof: the polar of the four-item linear-ordering polytope has a triangle-free Pfaffian interaction graph. The graph bound also yields the complete equality family below. The contribution here is the proof mechanism and equality description, rather than the numerical constant or its universal validity.

Theorem 5.1. *Every nondegenerate simplex $K \subset \mathbb{R}^4$ satisfies*

$$\frac{c_{\mathrm{EHZ}}(K)^2}{2\operatorname{vol}_4 K} \le \frac{3}{4}.$$

The constant is sharp. In interleaved symplectic coordinates $\omega_0 = dz_1 \wedge dz_2 + dz_3 \wedge dz_4$, equality holds precisely for the images under translations, positive dilations and linear symplectomorphisms of

$$K_{\mu,\nu} = \operatorname{conv}\{0, (1,0,\nu,-\nu),\ (0,1,-\mu,\mu),\ e_3,\ e_4\}, \qquad \mu,\nu \ge 0, \quad \mu+\nu \le 1.$$

5.1 Barycentric gradients remove the weight optimization

Let $\lambda_0, \dots, \lambda_4$ be the barycentric coordinate functions of K, and write $x_i = -d\lambda_i$ for their constant negative gradients, identified with vectors using the ambient Euclidean coordinates. They obey $\sum_{i=0}^4 x_i = 0$, and any four form a basis. Translate K so that the origin is interior, and set $\beta_i = \lambda_i(0) > 0$. Its normalized facet presentation is

$$K = \{z : a_i \cdot z \le 1,\ 0 \le i \le 4\}, \qquad a_i = x_i/\beta_i.$$

The unique normalized nonnegative closure vector in Haim–Kislev's formula is (β_i): the kernel of the matrix of the five facet rows is one-dimensional, and $\sum_i \beta_i = 1$, $\sum_i \beta_i a_i = 0$.

Define the alternating matrix

$$\Omega_{ij} = \omega_0(x_i, x_j), \qquad 1 \le i, j \le 4,$$

and the order functional

$$S(\Omega) = \max_{\sigma \in \mathfrak{S}_4} \sum_{i<j} \Omega_{\sigma(i),\sigma(j)}. \tag{24}$$

Closure makes cyclic rotation preserve the ordered pairing sum. Put x_0 last; its contribution is $\omega_0(\sum_{i=1}^4 x_i, x_0) = \omega_0(-x_0, x_0) = 0$. The Haim–Kislev formula [6, Theorem 1.1] therefore gives

$$c_{\mathrm{EHZ}}(K) = \frac{1}{2S(\Omega)}, \qquad \operatorname{vol}_4 K = \frac{1}{24|\operatorname{Pf}\Omega|}. \tag{25}$$

For the second identity, the affine map $z \mapsto (\lambda_1(z), \dots, \lambda_4(z))$ takes K onto the standard simplex of volume $1/24$. Its absolute Jacobian is the absolute determinant of the four gradients. The Pfaffian congruence identity gives the same absolute determinant for Ω. Thus

$$\frac{c_{\mathrm{EHZ}}(K)^2}{2\operatorname{vol}_4 K} = \frac{3|\operatorname{Pf}\Omega|}{S(\Omega)^2}. \tag{26}$$

Conversely, every nondegenerate alternating matrix is a symplectic Gram matrix of a basis, so this reduction describes the whole affine simplex class. The gradients determine the labeled simplex up to translation; equal Gram matrices identify them up to an affine symplectomorphism.

5.2 The polar of the four-item ordering polytope

Use coordinates $(12, 13, 14, 23, 24, 34)$ on the six-dimensional space of alternating matrices, and let e_{ij} denote its coordinate unit vectors. In these coordinates,

$$q(z) := \operatorname{Pf} z = z_{12}z_{34} - z_{13}z_{24} + z_{14}z_{23}.$$

For $\sigma \in \mathfrak{S}_4$, let p_σ have coordinate $+1$ on ij when i precedes j and -1 otherwise. Then S is the support function of $P = \operatorname{conv}\{p_\sigma\}$.

We use the classical facet description of the four-item linear-ordering polytope: in incidence coordinates $y = (1+p)/2$, it is defined by

$$0 \le y_{ij} \le 1, \qquad 0 \le y_{ij} - y_{ik} + y_{jk} \le 1 \quad (i < j < k).$$

After centering and polarity this says

$$C := \{z : S(z) \le 1\} = \operatorname{conv}\mathcal{V}, \quad \mathcal{V} = \{\pm e_{ij} : i < j\} \cup \{\pm(e_{ij} - e_{ik} + e_{jk}) : i < j < k\}. \tag{27}$$

There are twenty vertices. This small classical polytope identity is the only ordering-polytope input needed below.

For completeness, the facet description has a short integrality proof. Let R be the polyhedron defined by the displayed bounds and triangle inequalities. At any vertex of R, some coordinate bound is active: otherwise at most one constraint from each of the four triangle pairs can be active, fewer than the six independent constraints needed at a vertex. Relabeling the four items, we may suppose $y_{12} = 1$. Write $(a, b, c, d, e) = (y_{13}, y_{14}, y_{23}, y_{24}, y_{34})$. On this face the remaining constraints are the unit box and

$$c \le a, \quad d \le b, \quad 0 \le a - b + e \le 1, \quad 0 \le c - d + e \le 1.$$

The four core row vectors are

$$\begin{pmatrix} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 1 \end{pmatrix}.$$

Changing their signs by $(+, -, -, +)$ gives a matrix with one $+1$ and one -1 in every column. Every square minor of such a directed-incidence matrix is 0 or ± 1: expand along a column

with at most one nonzero entry, or, if every column has two, use the zero sum of the rows. Appending coordinate rows and signed copies preserves this property. An invertible active constraint matrix consequently has determinant ± 1, and its integer right-hand side forces the vertex coordinates to be integers. Thus every vertex of R is a zero–one vector. The triangle inequalities exclude directed triangles; a tournament without directed triangles is a total order. Hence R is exactly the convex hull of the twenty-four order vectors, proving (27) without a computational assumption.

5.3 A triangle-free Pfaffian interaction graph

Let $f_1, \dots, f_4$ be the standard basis of $\mathbb{R}^4$ and $f_5 = f_1 + \cdots + f_4$. With the usual Hodge star,

$$\mathcal{V} = \{\pm * (f_i \wedge f_j) : 1 \le i < j \le 5\}.$$

The six pairs not involving f_5 give the coordinate vectors; the other four give the triangle vectors. Every vertex is decomposable and hence has $q(v) = 0$. Define its symmetric interaction with another vertex by

$$B(v, w) = q(v + w) - q(v) - q(w).$$

If $v = \varepsilon * (f_i \wedge f_j)$ and $w = \eta * (f_k \wedge f_l)$, then

$$B(v, w) = \varepsilon\eta \det(f_i, f_j, f_k, f_l).$$

This is zero when the two index pairs intersect, and is $+1$ or -1 otherwise: every four distinct f_i have determinant ± 1.

Let G have vertex set $\mathcal{V}$, with vw an edge exactly when $B(v, w) = 1$. It is triangle-free, since three pairwise disjoint two-element subsets cannot fit inside a five-element set. It is also cubic: a pair has three disjoint pairs, and exactly one sign of each gives positive interaction. These elementary facts already prove the bound.

Lemma 5.2 (Triangle-free weighted graph bound)**.** *If G is triangle-free and $\lambda_v \ge 0$, $\sum_v \lambda_v = 1$, then*

$$\sum_{vw \in E(G)} \lambda_v \lambda_w \le \frac{1}{4}.$$

Proof. Choose an optimizing weight vector with the fewest positive entries. If two supported vertices are nonadjacent, redistribute their combined mass to the one with larger weighted neighbor sum. The objective is linear in this redistribution, so it does not decrease, contradicting minimal support. The support is therefore a clique, of size at most two. Its edge contribution is at most $t(1 - t) \le 1/4$. □

This is the triangle-free case of the Motzkin–Straus theorem [11, Theorem 1]; its short proof is included to make the numerical constant transparent.

Proof of Theorem 5.1. For $z \in C$, write $z = \sum_v \lambda_v v$ as a convex combination of vertices in (27). Since $q(v) = 0$,

$$q(z) = \sum_{v<w} \lambda_v \lambda_w B(v, w) \le \sum_{vw \in E(G)} \lambda_v \lambda_w \le \frac{1}{4}.$$

An odd permutation of the four coordinate indices preserves S and reverses the Pfaffian sign. Applying the same argument after such a permutation proves $|q(z)| \le 1/4$ on C. Homogeneity gives $4|q(\Omega)| \le S(\Omega)^2$; now use (26). The standard simplex has $c_{\mathrm{EHZ}} = 1/4$ and volume $1/24$, giving equality. □

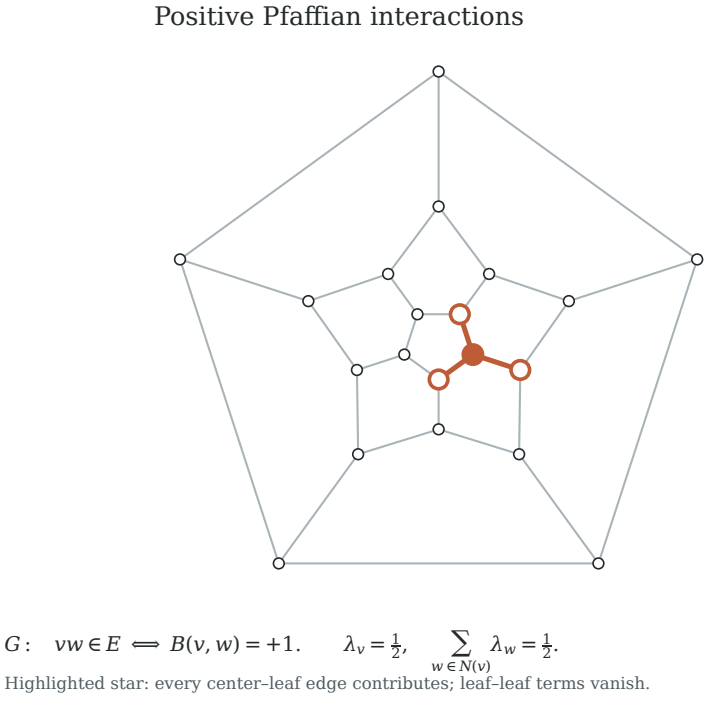


Figure 2: The positive-interaction graph on signed rank-two vertices. The highlighted star shows an equality support: half the mass is at its center and half is distributed among its three neighbors.

5.4 All equality forms

Two distinct vertices of G have at most one common neighbor. To see this, forget signs and label a vertex by its two-element subset of $\{1, \dots, 5\}$. Two distinct subsets have at most one common disjoint pair. If the signed vertices have the same underlying subset, they have opposite signs and no common positive neighbor. Thus G has no four-cycle. Its unsigned quotient is the Petersen graph. Only the established absence of triangles and four-cycles is used below.

Proposition 5.3. *The set* $\{z : S(z) = 1,\ q(z) = 1/4\}$ *is exactly*

$$\bigcup_{v \in \mathcal{V}} \left(\frac{1}{2} v + \frac{1}{2} \operatorname{conv} N_G(v) \right).$$

Each $N_G(v)$ *consists of three vertices, so this is a union of twenty closed triangles. Negative-Pfaffian equality is obtained by any odd coordinate permutation, and positive scaling gives all nonzero equality forms.*

Proof. Suppose $q(z) = 1/4$ and choose a convex decomposition $z = \sum_v \lambda_v v$. Both inequalities in the preceding proof are equalities. The weights maximize the graph quadratic at $1/4$. On their positive support, first-order variations show that all weighted neighbor sums are equal. Their weighted average is twice the objective, so each is $1/2$.

Choose a supported edge uv. The supported neighbor sets of u and v are disjoint, by triangle-freeness, and each has mass $1/2$. They therefore partition the support. Each side is independent. Every vertex must be adjacent to the whole opposite side, since its weighted degree is $1/2$ and every supported weight is positive. The support is complete bipartite.

The absence of four-cycles forces one side to be a singleton: the support is a star, its center has mass $1/2$, and its leaves have total mass $1/2$.

Conversely, any two neighbors of a vertex have underlying pairs in the same remaining three-element set. They intersect, so their interaction B is zero. Every such balanced star therefore has Pfaffian $1/4$. It lies in C; the proved inequality forces $S = 1$. □

For example, the positive neighbors of e_{12} are

$$e_{34}, \qquad e_{13} - e_{14} + e_{34}, \qquad e_{23} - e_{24} + e_{34}.$$

Doubling this equality triangle gives the explicit family

$$\Omega = e_{12} + e_{34} + \mu(e_{13} - e_{14}) + \nu(e_{23} - e_{24}), \qquad \mu, \nu \geq 0, \quad \mu + \nu \leq 1,$$

with $q(\Omega) = 1$ and $S(\Omega) = 2$. The union in the proposition is the complete equality classification in pairing coordinates.

To pass from twenty triangles to the family in the theorem, relabel the five barycentric gradients. If B_π expresses the new four-vector basis in the old one, the induced action is $\Omega \mapsto B_\pi^T \Omega B_\pi$. It preserves S, and $\det B_\pi = \operatorname{sgn} \pi$. An odd relabeling first makes the Pfaffian positive. The even relabelings act transitively on $\mathcal{V}$: even permutations fixing the omitted label give all twelve signed coordinate vectors from e_{12}; exchanging the omitted label with a third index, and exchanging the first two indices to preserve parity, gives the eight signed triangle vectors. They therefore take any equality triangle to the displayed one. Scaling then sets $q = 1$.

For an explicit realization, let X have columns

$$e_1, \ e_2, \ -\nu e_1 + \mu e_2 + e_3, \ \nu e_1 - \mu e_2 + e_4.$$

Then $X^T J X$ is the displayed Ω, and X^{-T} has columns $(1, 0, \nu, -\nu)$, $(0, 1, -\mu, \mu)$, e_3, e_4. The negative barycentric gradients of $K_{\mu,\nu}$ are the negatives of the columns of X, together with their opposite sum. Hence this simplex has the required pairing matrix. It has volume $1/24$ and capacity $1/4$. Equal pairing matrices give affine symplectomorphisms, so every equality simplex has the stated form up to the claimed transformations.

The parameter boundary has a geometric meaning. The ten pairings of the five weighted facet normals have absolute values $1, 1, 1, 1, \mu, \mu, \nu, \nu, 1 - \mu - \nu, 1 - \mu - \nu$. An intersection of two facets is Lagrangian exactly when their normal pairing vanishes. Thus interior parameters give equality simplices with no Lagrangian two-faces, open edges give two, and the three corners give four. Thus the equality family contains simplices with no Lagrangian two-faces, as observed experimentally by Chaidez and Hutchings.

6 A six-dimensional simplex and a hypergraph Lagrangian

In dimension six, the standard simplex no longer maximizes the systolic ratio, as Haim–Kislev observed numerically in her 2019 thesis [5, Remark 1.3]. The pairing formulation gives a particularly simple exact example and explains a numerical optimizer through a classical hypergraph Lagrangian. We prove that the example is a strict local maximum among simplices, modulo the natural symmetries; Section 7 proves it is globally optimal and classifies equality.

For an alternating 6×6 matrix Ω, put

$$S_6(\Omega) = \max_{\sigma \in \mathfrak{S}_6} \sum_{i<j} \Omega_{\sigma(i),\sigma(j)}, \qquad C_6 = \{\Omega : S_6(\Omega) \leq 1\}.$$

The same barycentric-gradient argument as in the preceding section gives

$$\frac{c_{\mathrm{EHZ}}(K)^3}{6\,\mathrm{vol}_6\,K} = \frac{15|\,\mathrm{Pf}\,\Omega|}{S_6(\Omega)^3} \tag{28}$$

for every six-simplex. Indeed the capacity is $1/(2S_6)$ and the volume is $1/(6!|\,\mathrm{Pf}\,\Omega|)$. Every nondegenerate alternating Ω occurs, and identical pairings determine labeled simplices up to affine symplectomorphism.

Write e_{ij} for an alternating coordinate vector and $t_{ijk} = e_{ij} - e_{ik} + e_{jk}$ for $i < j < k$. Consider the five forms

$$v_1 = t_{135}, \qquad v_2 = -t_{246}, \qquad v_3 = e_{12}, \qquad v_4 = e_{34}, \qquad v_5 = e_{56}. \tag{29}$$

Each has $S_6(v_i) = 1$: a coordinate order score is ± 1, as is the score on an oriented triangle. The order $(1, 3, 5, 2, 6, 4)$ has score 1 on all five. Consequently $S_6 = 1$ on their entire convex hull.

Theorem 6.1. *Let*

$$a = \frac{13 + 3\sqrt{5}}{62}, \qquad b = \frac{6 - \sqrt{5}}{31}, \qquad \Omega_* = a(t_{135} - t_{246}) + b(e_{12} + e_{34} + e_{56}).$$

Then $S_6(\Omega_*) = 1$ *and*

$$\mathrm{Pf}\,\Omega_* = \frac{63 + 5\sqrt{5}}{1922}.$$

The corresponding six-simplex has systolic ratio

$$\frac{945 + 75\sqrt{5}}{1922} > \frac{5}{9},$$

strictly exceeding the standard simplex's ratio. It is a strict local maximum among six-simplices modulo translations, positive dilations and linear symplectomorphisms.

The value is approximately 0.5789308524; its cube root is approximately 0.8334423504. Leipold reports the nearby root-normalized numerical value 0.8334409621 [9, Section 12.4 and Appendix]. Numerical proximity alone does not identify the two simplex configurations. The exact proof here establishes this candidate's global optimality below.

6.1 The five-vertex face and its cubic

Direct Pfaffian expansion gives

$$\mathrm{Pf}(xv_1 + yv_2 + zv_3 + uv_4 + vv_5) = xy(z + u + v) + zuv. \tag{30}$$

Here the scalar variable v in the right-hand side is distinct from the indexed forms v_i. This is the edge polynomial of the 3-uniform hypergraph

$$F_{3,2} = \{123, 124, 125, 345\}.$$

Its Lagrangian and maximizing weights are classical: Baber and Talbot record exactly these weights and the value $(189 + 15\sqrt{5})/961$ with their normalization, which is six times the polynomial in (30) [1, Theorem 22]. Thus the symplectic ratio in Theorem 6.1 is 5/2 times that hypergraph Lagrangian. We include the short maximization argument to keep the geometric result independent of any extremal hypergraph theorem.

For nonnegative weights summing to one, put $t = z + u + v$. The arithmetic– geometric mean inequality gives

$$xy(z+u+v) + zuv \le \frac{(1-t)^2 t}{4} + \frac{t^3}{27} =: G(t), \qquad 0 \le t \le 1.$$

Equality requires $x = y = (1-t)/2$ and $z = u = v = t/3$, apart from irrelevant zero-mass endpoint cases. Comparing the endpoints and the two critical points of

$$G(t) = \frac{t}{4} - \frac{t^2}{2} + \frac{31t^3}{108}$$

gives its unique maximum at $t_* = (18 - 3\sqrt{5})/31$. This is precisely the stated pair of weights a, b. Moreover the restricted Hessian is negative definite. To see this, write $x = (1-t)/2 + \delta$, $y = (1-t)/2 - \delta$, and $(z, u, v) = (t/3 + \eta_1, t/3 + \eta_2, t/3 + \eta_3)$ with $\sum \eta_i = 0$. The polynomial becomes

$$G(t) - t\delta^2 - \frac{t}{6}\sum_i \eta_i^2 + \eta_1\eta_2\eta_3.$$

At $t = t_*$, $G''(t_*) = -\sqrt{5}/6 < 0$; the other quadratic terms are strictly negative on the three remaining tangent directions.

We next verify that this weight simplex is an actual face of the full fifteen-dimensional polytope C_6. This point matters: optimization on five chosen generators alone would not imply local maximality under arbitrary affine deformations.

For an order σ, let p_σ be its fifteen-coordinate sign vector. Let $\tau = (1\,3\,5)(2\,4\,6)$ act on the labels of an order. Let $\mathcal{I}$ consist of the three cyclic images under τ of each of the following seven orders:

$$\begin{array}{llll} 123564, & 132564, & 134256, & 134526, \\ 135264, & 135426, & 135642. & \end{array} \tag{31}$$

These are exactly the twenty-one orders having score 1 on all five forms in (29). Their common affine equality set $p_\sigma \cdot \Omega = 1$ is four-dimensional. More explicitly, subtracting the displayed equations gives

$$\begin{gathered} \Omega_{14} = \Omega_{16} = \Omega_{23} = \Omega_{25} = \Omega_{36} = \Omega_{45} = 0, \\ \Omega_{13} = \Omega_{35} = -\Omega_{15}, \qquad \Omega_{24} = \Omega_{46} = -\Omega_{26}, \\ \Omega_{13} - \Omega_{24} + \Omega_{12} + \Omega_{34} + \Omega_{56} = 1. \end{gathered}$$

These are eleven independent affine equations and identify the equality set with $\sum_{i=1}^5 \lambda_i v_i$, $\sum_i \lambda_i = 1$. For $i = 1, \dots, 5$, respectively, the orders

$$125364, \quad 123456, \quad 213564, \quad 142356, \quad 123645$$

have score -1 on v_i and score 1 on the other four forms. Their inequalities read $1 - 2\lambda_i \le 1$, forcing $\lambda_i \ge 0$. It follows that the common face is exactly $F = \operatorname{conv}\{v_1, \dots, v_5\}$.

6.2 A certificate against every transverse affine deformation

Set $q(\Omega) = \operatorname{Pf}\Omega$. Let $\mathcal{J}$ be the three cyclic images of the distinguished order 135264 in (31). Differentiating the Pfaffian at Ω_* gives the exact identity

$$\nabla q(\Omega_*) = \alpha \sum_{\sigma \in \mathcal{I} \setminus \mathcal{J}} p_\sigma + \beta \sum_{\sigma \in \mathcal{J}} p_\sigma, \quad \alpha = \frac{41 - 12\sqrt{5}}{5766} > 0, \quad \beta = \frac{-19 + 29\sqrt{5}}{1922} > 0. \tag{32}$$

This is a fifteen-coordinate identity over $\mathbb{Q}(\sqrt{5})$; the explicit orders and coefficients give a small independently checkable certificate.

For every feasible displacement h from Ω_*, each active order satisfies $p_\sigma \cdot h \le 0$. The positivity in (32) implies $\nabla q(\Omega_*) \cdot h \le 0$, with equality precisely when h lies in the tangent space of F. On that tangent space, the Hessian is negative definite by the preceding computation.

These two facts imply strict local maximality on C_6. Indeed, if nonzero feasible displacements $h_k \to 0$ satisfied $q(\Omega_* + h_k) \ge q(\Omega_*)$, pass to a convergent subsequence of $h_k/\|h_k\|$, with unit limit d. Taylor expansion first forces $\nabla q(\Omega_*) \cdot d = 0$, so d belongs to the face tangent space. The Hessian is strictly negative in direction d. The linear term is nonpositive for every h_k, so the second-order Taylor expansion now gives a strict decrease for large k, a contradiction.

Finally, radial normalization by S_6 identifies nearby nondegenerate pairing forms, modulo dilation, with a neighborhood in the boundary of C_6. Formula (28) transfers the strict local Pfaffian maximum to the stated strict local systolic maximum, modulo affine symplectomorphisms and translations. This proves Theorem 6.1.

The local certificate uses no complete facet enumeration of the linear-ordering polytope. The global proof in the next section uses an exact computer-assisted vertex classification. The contrast with dimension four is structural: a quadratic graph optimization is replaced by a cubic hypergraph optimization, and the hypergraph $F_{3,2}$ already appears using only coordinate and triangle forms.

7 The sharp global six-simplex inequality

The local example from Theorem 6.1 is in fact optimal among all six-simplices. The proof combines a small hypergraph argument with an exact finite description of a linear-ordering relaxation. The finite description is independently reproduced with two exact polyhedral backends; the inequalities used after that description are explicit.

Theorem 7.1. *Every six-dimensional simplex K satisfies*

$$\frac{c_{\mathrm{EHZ}}(K)^3}{6\,\mathrm{vol}_6 K} \le \frac{945 + 75\sqrt{5}}{1922}.$$

Equality holds precisely for the simplex in Theorem 6.1, up to translation, positive homothety, and linear symplectomorphism.

Put $q_* = (63 + 5\sqrt{5})/1922$. By (28), it suffices to maximize $|\operatorname{Pf}\Omega|$ on C_6. Let

$$D_6 = \operatorname{conv}\big(\{\pm e_{ij} : 1 \le i < j \le 6\} \cup \{\pm t_{ijk} : 1 \le i < j < k \le 6\}\big),$$

so D_6 has seventy rank-two generators and is contained in C_6.

7.1 The rank-two hull

Introduce seven points $u_1, \dots, u_6, u_7$ with $u_i = e_i$ for $i \le 6$ and $u_7 = 0$. For each three-element set $T = \{i, j, k\}$, choose the form

$$b_T = (u_i - u_k) \wedge (u_j - u_k).$$

Then $D_6 = \operatorname{conv}\{\pm b_T\}$. The orientation choices will not matter for an upper bound. If two triples share two labels, their forms share a wedge factor, so $b_T \wedge b_U = 0$. Also $b_T \wedge b_T = 0$.

Consequently the Pfaffian cubic in coefficients of these forms is multiaffine, and contains no monomial involving both T and U when $|T \cap U| \geq 2$.

Cancel antipodal coefficients in a convex decomposition. It becomes $\Omega = \sum_T d_T b_T$ with $\sum_T |d_T| \leq 1$. Homogeneity permits a nonzero global maximizer to be chosen with total mass one. Among global maximizers choose one with smallest support. If two supported types share two labels, fix their signs and their total absolute mass, and vary the split between them. The Pfaffian is affine in that split. Its absolute value therefore attains a maximum at an endpoint, removing one supported type without lowering the objective. This contradicts minimality. Thus some global maximum is supported on a family of triples whose pairwise intersections have size at most one.

We record the elementary classification needed here. A maximal such family on seven labels has either five or seven triples. Indeed, each triple uses three of the twenty-one label pairs, so there are at most seven triples. No label can have degree zero: otherwise maximality requires all pairs on the other six labels to be covered, which would give degree $5/2$ at each of those labels. If a label has degree one, with its only triple vab, all six pairs among the remaining four labels R must be covered. Without a triple contained in R, this requires six additional triples, forcing all twenty-one pairs to be covered and contradicting degree one. With a triple $xyz \subset R$, the three pairs wx, wy, wz, where $R = \{w, x, y, z\}$, require three separate triples using a or b. One of the pairs wa, wb would then repeat. Hence every label has degree at least two, and there are at least five triples. Six triples would leave three uncovered pairs with even degree at every label, hence an uncovered triangle, contradicting maximality.

With five triples the degree sequence is $(3, 2, 2, 2, 2, 2, 2)$. The three triples through the degree-three label match the other six labels. The two remaining triples partition those six labels, and each matching pair crosses that partition. Up to relabeling this is

$$\{135, 246, 127, 347, 567\}.$$

With seven triples every label pair is covered exactly once: the family is the Fano plane.

For three members of a linear triple family, the Pfaffian coefficient is zero if their union has at most six labels, since the forms lie in a difference space of dimension at most five. When their union has seven labels, their six edge factors form a spanning tree and the coefficient is ± 1. For the five-triple configuration, taking absolute values therefore gives the polynomial

$$xy(z + u + v) + zuv.$$

Its maximum under nonnegative total mass one is q_*, as computed in (30). For the Fano configuration the unsigned polynomial is the Fano-plane Lagrangian, because the nonzero triples are precisely three concurrent Fano lines.

Here is a short bound for the latter polynomial P. On the boundary, remove a zero-weight point. The four remaining Fano triples are transversal to a partition of the six remaining points into three pairs. Hence P is at most the product of the three class masses, and is at most $1/27$. At a full-support maximum, the seven partial derivatives all equal $3P$. Their sum counts every pair exactly once, giving

$$21P = \sum_{i<j} x_i x_j \leq \frac{3}{7}, \qquad P \leq \frac{1}{49}.$$

Thus $P \leq 1/27 < q_*$. Together with the explicit attaining example, this proves

$$\max_{\Omega \in D_6} |\operatorname{Pf} \Omega| = q_*. \tag{33}$$

7.2 The finite polyhedral decomposition

All pairings in this subsection use the fifteen upper-triangular entries. Define

$$F = \frac{1}{5}\begin{pmatrix} 0 & M \\ -M^T & 0 \end{pmatrix}, \quad M = 2I - \mathbf{1}\mathbf{1}^T, \qquad p_0 = \begin{pmatrix} 0 & N \\ -N^T & 0 \end{pmatrix}, \quad N = I - \mathbf{1}\mathbf{1}^T.$$

The form F is a normalized three-fence. Let $\mathcal{F}$ be its orbit under the seven-label permutation action. The action on forms comes from relabeling the seven barycentric gradients and has determinant ± 1; it preserves C_6, D_6 and the absolute Pfaffian. Its dual action on functionals is contragredient.

The retained replay in `certificates/sixdim-global/README.md` specifies four small programs:

`algebra.py`	`local_maximum.py`
`full_classification.py`	`dual_coverage.py`

The latter two use exact cdd and PPL polyhedral backends. Their generated outputs and a fresh copied-package run receipt are retained beside the programs. This is a computer-assisted completeness check, not a formalized proof in a proof assistant.

The exact finite calculations establish the following facts:

1. $Q_6 = D_6^\circ$ has exactly 1560 vertices: the 720 order functionals and the 840 elements of the dual orbit of p_0.

2. $\mathcal{F}$ has 840 elements. The canonical pairing is $p_0 \cdot F = 6/5$, whereas $p_0 \cdot F' \le 4/5$ for every other $F' \in \mathcal{F}$. Every fence satisfies every order inequality.

3. The vertex p_0 has eighteen neighbors in Q_6, all order functionals. They are exactly the eighteen orders with $F \cdot p = 1$. The face $B = \{X \in D_6 : p_0 \cdot X = 1\}$ has eighteen rank-two vertices and affine dimension fourteen.

For completeness, the completeness check in the first item uses graph connectivity, not a sample of vertices. Six adjacent-transposition generators preserve the seventy inequalities of Q_6 and generate the two disjoint orbits. At an order representative, both exact polyhedral backends give 186 extreme tangent rays, whose exact endpoints are 165 order vertices and 21 fractional vertices. At p_0 they give eighteen rays, all with order endpoints. Both representatives have active rank fifteen. The candidate set is therefore closed under every neighbor in the one-skeleton. Since a bounded polytope has connected one-skeleton, there are no further vertices.

Cut Q_6 by the 840 inequalities $F' \cdot p \le 1$. Each cut removes its corresponding fractional vertex and meets all its incident edges at existing integral neighbors. This can be done sequentially without creating vertices: all other fractional vertices lie strictly inside the current cut, so their tangent cones are unchanged; their original edges to integral neighbors remain feasible in full. Thus each later cut has the same crossing edges and introduces no new endpoint. The resulting polytope is precisely the convex hull of the order functionals. Polarity gives

$$C_6 = \operatorname{conv}(D_6 \cup \mathcal{F}).$$

Moreover, the midpoint of any two distinct fences lies in D_6. On an order functional both endpoint scores are at most one; on a fractional functional at most one is $6/5$, while the other is at most $4/5$. Replacing equal masses of two fences by their midpoint repeatedly reduces any convex combination to a point of D_6 or a combination of D_6 and one fence. Each fence violates exactly one facet inequality of D_6. Consequently C_6 is covered by D_6 and the 840 pyramids $\operatorname{conv}(F', B_{F'})$. It remains to bound one pyramid.

7.3 A quadratic bound on a fence pyramid

Write $X = \sum_{i=1}^{18} x_i v_i \in B$, with $x_i \geq 0$ and $\sum_i x_i = 1$. Exact Pfaffian expansion gives

$$\operatorname{Pf}(tF + sX) = \frac{4}{125}t^3 + \frac{2}{25}t^2 s + Q(X)ts^2 + \operatorname{Pf}(X)s^3.$$

Here $Q(X) = x^T Hx/10$, where H is a symmetric integer matrix with zero diagonal and entries $0, 1, 2$. The graph of entries equal to two is the disjoint union of six three-vertex paths. In each path, the two leaves have entry zero between them; entries between distinct paths are zero or one. These explicit coefficient properties give

$$\mathbf{1}\mathbf{1}^T - H = \sum_{\text{six paths } (c,l,r)} (e_c - e_l - e_r)(e_c - e_l - e_r)^T + R, \qquad R \geq 0 \text{ entrywise.}$$

Hence $0 \leq Q(X) \leq 1/10$. By (33), for $t, s \geq 0$,

$$|\operatorname{Pf}(tF + sX)| \leq \frac{4}{125}t^3 + \frac{2}{25}t^2 s + \frac{1}{10}ts^2 + q_* s^3 \leq q_*(t + s)^3.$$

The last inequality is strict when $t > 0$, because $q_* > 1/30$. This proves the global bound, with strictness outside D_6.

7.4 Equality

Begin with any global maximizer in D_6 and apply the signed mass-merging procedure. At each interior split, the absolute value of an affine function already has its global maximum; that affine function must be constant. Thus the procedure gives a finite path of global maximizers to a supported linear triple packing. The Fano case is too small. In the five-triple case, equality forces the golden weights and common signs for the two central variables and for the three outer variables. All four resulting sign patterns are relabelings of Ω_*: $(1\,3)(2\,4)$ flips the two triangle signs, and $(1\,2)(3\,4)(5\,6)$ negates the entire form.

Every endpoint is therefore a strict local maximum on C_6, by Theorem 6.1. A nonconstant segment of global maximizers cannot end at such a point. Working backward through the finite merging path shows that the original form was the same endpoint. Thus every maximizer is a seven-label relabeling of Ω_*. Identical normalized barycentric-gradient pairings determine simplices up to affine symplectomorphism, proving the equality statement in Theorem 7.1.

8 A sharp difference-area inequality for squared bodies

Let $\sigma : \mathbb{C} \to \mathbb{C}$ be $\sigma(z) = z^2$. We prove

$$\operatorname{area}(\sigma K - \sigma K) \leq 5 \operatorname{area}(\sigma K)$$

for every origin-symmetric planar convex body K, with a sharp constant. The proof identifies squared bodies with planar reciprocal bodies, reduces arbitrary sources to parallelograms, and bounds their squares by a geometric argument valid in every dimension. The last argument gives a sharp supremum for intersections of two confocal paraboloids; it does not bound arbitrary higher-dimensional reciprocal bodies.

For a planar body L write

$$R(L) = \frac{\operatorname{area}(L - L)}{4 \operatorname{area}(L)}.$$

This quotient is invariant under invertible affine maps. The planar bound is $R \le 5/4$. Through the quaternionic construction of Chaidez–Tanny [4], it bounds $c_{\mathrm{EHZ}}(X)^2$ by $(5/2)\,\mathrm{Vol}_4(X)$ for bodies with right quaternionic I-symmetry. We give the Finsler and capacity normalizations at the end of the section. A *source body* below is compact, origin-symmetric, convex, and has nonempty interior.

8.1 Complex squaring and reciprocal convexity

Following Milman–Milman–Rotem [10], write

$$M' = \{y : \langle y, n\rangle \le h_M(n)^{-1} \text{ for all } n \in S^1\}.$$

Their Lemma 18 and Proposition 19 express M' as the intersection, over $x \in M$, of the parabolic regions

$$P_x = \{y : \langle y, x\rangle \le 1 - \tfrac{1}{4}|x|^2 |\operatorname{proj}_{x^\perp} y|^2\}, \quad P_0 = \mathbb{R}^2.$$

For $x = re^{i\theta}$, elementary squaring gives

$$P_x = \sigma(S_x), \qquad S_x = \{z : |\operatorname{Re}(e^{-i\theta/2} z)| \le r^{-1/2}\}.$$

Indeed, in coordinates $e^{-i\theta}y = u + iv$, both descriptions are $u \le r^{-1} - rv^2/4$. The map σ commutes with arbitrary intersections of centrally symmetric sets: the two square roots of a point either both belong to any such set or both fail to belong.

Theorem 8.1 (Squared bodies and reciprocity)**.** *The compact planar reciprocal bodies containing the origin in their interior are precisely the images $\sigma(K)$ of source bodies. Moreover,*

$$(\sigma K)' = \sigma(K^\circ).$$

Proof. Write K as the intersection of its symmetric supporting strips with unit normals $n = e^{i\phi}$ and half-widths $h_K(n)$. Each strip maps to P_{x_ϕ}, where $x_\phi = e^{2i\phi}/h_K(n)^2$. Consequently σK is convex. Since $n/h_K(n)$ runs over ∂K°, the points x_ϕ run over $\partial\sigma(K^\circ)$. This last image is also convex by the same strip argument. The parabolic representation therefore gives

$$\sigma K = (\sigma(K^\circ))'.$$

Apply the same identity to K° to obtain the stated conjugacy. Conversely, the parabolic representation of any reciprocal M' identifies it with $\sigma(\bigcap_{x\in M} S_x)$. The intersection is centrally symmetric and convex; compactness and interior of its square imply the corresponding properties of the intersection itself. □

This is an identity in fixed Euclidean coordinates. Neither squaring nor reciprocity is asserted to commute with arbitrary affine maps.

8.2 Two confocal paraboloids in arbitrary dimension

We first record the shadow-system fact used twice below. If points move parallel to a fixed vector e according to

$$C_t = \operatorname{conv}\{p + t\beta(p)e : p \in S\},$$

then $\mathrm{Vol}_d(C_t)$ is convex in t. Indeed, the projection onto $e^\perp$ is fixed. On each line parallel to e, the upper endpoint is a supremum of affine functions of t, and the lower endpoint an infimum: the admissible convex combinations are specified by their fixed transverse coordinate. The fiber length is therefore convex, and integration proves the assertion. If convex bodies C_t are given by such moving points, then $C_t - C_t$ is another shadow system, with generators $p - q$ and velocities $\beta(p) - \beta(q)$.

Theorem 8.2 (Two-paraboloid difference-volume bound)**.** *Let $d \geq 2$, and let $Q \subset \mathbb{R}^d$ be a bounded intersection of two solid paraboloids of revolution with a common focus. Then*

$$\frac{\mathrm{Vol}_d(Q-Q)}{\mathrm{Vol}_d(Q)} < C_d, \qquad C_d = \frac{2^{d+1}(d-1)+2}{d}.$$

The constant C_d is the supremum over this class, and is approached by an affine-normalized degeneration to the round parabolic cap

$$P_d = \{(u,z) : 0 \leq u \leq 1,\ |z|^2 \leq u\}.$$

Proof. Normalize the pair. Translate the common focus to the origin. The two inequalities are

$$|y| + a_i \cdot y \leq c_i, \qquad |a_i| = 1, \quad c_i > 0, \quad a_1 \neq a_2.$$

The last condition is necessary for boundedness. On the future light cone $r = |y|$, the covectors $\ell_i = (1, a_i)/c_i$ are null for the quadratic form $r^2 - |y|^2$. Their sum is timelike, since

$$(\ell_1 + \ell_2)_0^2 - |(\ell_1 + \ell_2)_{\mathrm{sp}}|^2 = \frac{2(1 - a_1 \cdot a_2)}{c_1 c_2} > 0.$$

A Lorentz transformation sends that sum to the time axis. The transformed null covectors then have opposite spatial parts and equal positive time coordinates. A common rescaling and a spatial rotation give the lens

$$L = \{(x,z) : \sqrt{x^2 + |z|^2} + |x| \leq 1\} = \{(x,z) : |x| \leq \tfrac{1}{2},\ |z|^2 \leq 1 - 2|x|\}.$$

This change preserves the class: Lorentz maps preserve the future cone and future null covectors, so each inequality remains a confocal paraboloid inequality after division by its positive time coefficient. A Lorentz transformation is a spatial rotation composed with a boost; the latter has spatial part $(y_1, y_\perp) \mapsto (\gamma(y_1 + t|y|), y_\perp)$, where $|t| < 1$ and $\gamma = (1-t^2)^{-1/2}$. Consequently, up to an invertible spatial linear map, every original Q is

$$T_v(L), \qquad T_v(y) = y + |y|v, \quad |v| < 1.$$

These images are convex by the same Lorentz calculation, and remain convex for $|v| = 1$ by Hausdorff continuity.

Enlarge the shadow family. Write $v = ca + b$, where a is the x-axis and $b \perp a$. Define

$$F_{c,b}(x,z) = (x,z) - |x|(c,b).$$

We claim that $F_{c,b}(L)$ is convex throughout the larger parameter set $|c| \leq 1$, $|b| \leq 1$. For $|c| < 1$, its positive and negative halves are affine images of the half-lenses, on opposite sides of the common interface $D = \{0\} \times B^{d-1}$. Take one point in either half:

$$Y_+ = ((1-c)t, z - tb), \qquad Y_- = (-(1+c)r, w - rb),$$

where $t, r \geq 0$, $|z|^2 \leq 1 - 2t$ and $|w|^2 \leq 1 - 2r$. The segment crosses the interface at a point with transverse coordinate

$$z_0 = \lambda z + (1-\lambda)w - [\lambda t + (1-\lambda)r]b$$

for some $\lambda \in [0,1]$. Since $\sqrt{1-2t} \leq 1 - t$ and likewise for r, we have $|z_0| \leq 1$. The crossing lies in D, so the whole segment lies in the union of the two convex halves. This proves convexity; the endpoint values of c follow by continuity.

On ∂L we have $|y| = 1 - |x|$, and hence $T_v(y) = v + F_{c,b}(y)$. For $|v| < 1$, the map T_v is a homeomorphism: on each preserved transverse line, its coordinate along v is strictly increasing. Thus support maxima of $T_v(L)$ occur on $T_v(\partial L)$. Support maxima of $F_{c,b}(L)$ occur on the two outer parabolic sheets or the interface rim, also images of points of ∂L. Equality of support functions gives

$$T_v(L) = v + F_{c,b}(L), \qquad |v| \le 1,$$

with the unit-vector case obtained by continuity. This is equality of image bodies; the maps need not agree at interior points.

The two affine half-maps have determinants $1 - c$ and $1 + c$. Their interiors are disjoint and the original half-volumes equal, so

$$\mathrm{Vol}_d(F_{c,b}(L)) = \mathrm{Vol}_d(L).$$

For fixed b, varying c moves all points parallel to a, with speed $-|x|$. The difference-body volume is therefore convex in c, and its quotient by volume is bounded by the larger quotient at $c = \pm 1$.

Identify the endpoints. At $c = 1$ the positive half collapses into D, because $|z - tb| \le |z| + t \le 1$. The negative half has coordinates

$$u = -2r, \qquad w = z - rb, \qquad |z|^2 \le 1 - 2r.$$

The affine change $U = 1 + u$, $Z = w - (u/2)b$ identifies it with P_d. The other endpoint is its reflection. Writing ω_{d-1} for the volume of the unit $(d-1)$-ball, we obtain

$$\mathrm{Vol}_d(P_d) = \frac{2\omega_{d-1}}{d+1}, \qquad \mathrm{Vol}_d(P_d - P_d) = 2\omega_{d-1} \int_0^1 (1 + \sqrt{1-s})^{d-1}\, ds.$$

For fixed difference coordinate $s \ge 0$, the transverse difference balls increase as the source coordinate increases; the largest is attained at the pair $1, 1 - s$. Evaluating the integral gives the stated quotient C_d.

Finally, along the original radial shadow $T_{te}(L)$ the volume is constant: its Jacobian is $1 + t\langle e, y\rangle/|y|$, and the odd term integrates to zero. The quotient at $t = 0$ is $2^d < C_d$, whereas that at $t = 1$ is at most C_d. Convexity of the numerator gives strictness for $|t| < 1$. As $v \to a$, the images tend to a translate of P_d; volume continuity proves sharpness of the supremum. □

8.3 Endpoint caps and positive tent decompositions

For a squared planar body L, let

$$T_t(u, v) = (u + t\sqrt{u^2 + v^2}, v), \qquad -1 \le t \le 1.$$

For $|t| < 1$, this is affine equivalent to another squared body, since

$$\sigma(\sqrt{1+t}\, x + i\sqrt{1-t}\, y) = (\mathrm{Re}\, \sigma z + t|\sigma z|, \sqrt{1-t^2}\, \mathrm{Im}\, \sigma z).$$

The images are convex, including the endpoints by continuity. The Jacobian gives

$$\mathrm{area}(T_t L) = A + tB, \qquad A = \mathrm{area}(L), \quad B = \int_L \frac{u}{\sqrt{u^2+v^2}}\, du\, dv, \qquad |B| < A.$$

The difference bodies form a shadow system, so their areas are convex in t. Dividing the endpoint interpolation by the positive affine function $4(A + tB)$ proves

$$R(L) \le \max\{R(T_{-1}L), R(T_1L)\}.$$

Since $T_1(\sigma(x + iy)) = (2x^2, 2xy)$, it suffices to bound the caps

$$C_K = \{(x^2, xy) : (x, y) \in K\}.$$

The other endpoint is the same construction after exchanging source coordinates.

Normalize the horizontal projection of K to $[-1, 1]$ and write

$$K = \{(x, y) : -1 \le x \le 1,\ -b(-x) \le y \le b(x)\},$$

where b is concave. A source shear $y \mapsto y+cx$ induces the affine cap shear $(u, v) \mapsto (u, v+cu)$. Choose it so that $b(-1) = b(1) = d \ge 0$; then $b \ge d$. The cap sections are

$$-V_b(u) \le v \le U_b(u), \quad U_b(u) = \sqrt{u}\, b(\sqrt{u}), \quad V_b(u) = \sqrt{u}\, b(-\sqrt{u}), \qquad 0 \le u \le 1.$$

Their area $A(b) = \int_0^1 (U_b + V_b)$ is linear in b. Define

$$H_+(b, s) = \max_{0\le u\le 1-s} [U_b(u + s) + V_b(u)],$$
$$H_-(b, s) = \max_{0\le u\le 1-s} [V_b(u + s) + U_b(u)].$$

The upper and lower endpoints of each difference section give

$$N(b) := \operatorname{area}(C_b - C_b) = 2\int_0^1 [H_+(b, s) + H_-(b, s)]\, ds.$$

Each maximum is positively homogeneous and convex in b; thus N is subadditive.

For $-1 < a < 1$, let

$$f_a(x) = \begin{cases} (1 + x)/(1 + a), & x \le a, \\ (1 - x)/(1 - a), & x \ge a. \end{cases}$$

Every continuous concave profile with equal endpoints is a uniform limit of positive combinations of 1 and these tents. For piecewise-linear profiles this follows by matching their slope jumps; uniform concave piecewise-linear approximation gives the general case. By linearity of A and subadditivity of N, it therefore suffices to prove $N \le 5A$ for the constant profile and for each single tent.

Theorem 8.3 (Sharp planar difference-area bound)**.** *For every origin-symmetric planar convex body K,*

$$\operatorname{area}(\sigma K - \sigma K) \le 5 \operatorname{area}(\sigma K).$$

The constant five cannot be decreased. The same inequality holds for all planar reciprocal bodies and their affine images.

Proof. The source of f_a is the parallelogram

$$K_a = \operatorname{conv}\{(-1, 0), (a, 1), (1, 0), (-a, -1)\};$$

the constant profile has a rectangular source. Each is an intersection of two symmetric strips, so its square is the intersection of two solid confocal parabolas. Any invertible flattening remains a parallelogram. Theorem 8.2 with $d = 2$ bounds the difference-area quotient of each

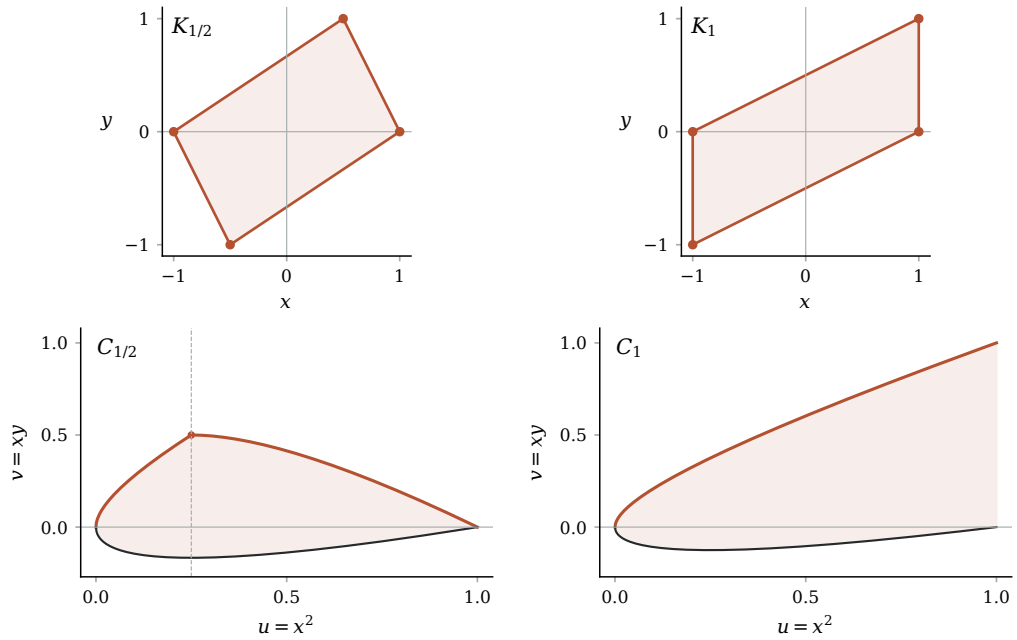


Figure 3: A single-tent source is a parallelogram. Its cap is a limit of affine-normalized squares of flattened parallelograms, so the two-paraboloid theorem applies without evaluating the difference-area integral.

flattened square by five. After affine normalization, these squares tend to the corresponding cap; positive limiting area and Hausdorff continuity pass the bound to that cap. The positive-profile reduction and then the endpoint reduction prove the inequality.

For sharpness, take $K_\epsilon = [-1, 1] \times [-\epsilon, \epsilon]$. Under $(u, v) \mapsto (u, v/(2\epsilon))$, its square tends to $P_2 = \{0 \le u \le 1,\ |v| \le \sqrt{u}\}$, whose difference area is five times its area. Reciprocity and affine invariance give the remaining statements. Lower-dimensional sources follow by approximation. □

8.4 Finsler and symplectic consequences

We give the normalization in the transfer explicitly. For a Finsler metric F on $\mathbb{RP}^2$, let $\ell(F)$ be its shortest noncontractible closed-geodesic length, and put

$$\mathcal{V}(F) = \int_{D_F^*\mathbb{RP}^2} \frac{\omega_{\mathrm{can}}^2}{2}.$$

This is the unnormalized cotangent volume used by Chaidez–Tanny; it is π times the usual normalized Holmes–Thompson area. Let $\overline{F}(v) = (F(v) + F(-v))/2$. Its unit codisks are $(D_x^*F - D_x^*F)/2$, because the support function of this Minkowski average is the average of the two tangent norms. The sharp planar theorem therefore gives

$$\mathcal{V}(\overline{F}) \le \frac{5}{4}\mathcal{V}(F)$$

whenever all the unit codisks of F are affine images of squared bodies.

Corollary 8.4. *For such a Finsler metric on* $\mathbb{RP}^2$,

$$\ell(F)^2 \le \frac{5}{8}\mathcal{V}(F).$$

Proof. For any noncontractible loop, at least one of its two orientations has F-length at most its $\overline{F}$-length. Thus $\ell(F) \le \ell(\overline{F})$. The reversible Ivanov–Pu inequality, in the normalization above, is $\ell(\overline{F})^2 \le \mathcal{V}(\overline{F})/2$; see [4, Theorem 3.7 and Lemma 3.8]. Combine these inequalities. □

Corollary 8.5. *Let $X \subset \mathbb{H}$ be a full-dimensional convex body invariant under right multiplication by the quaternion I, with the standard symplectic structure in [4]. Then*

$$c_{\mathrm{EHZ}}(X)^2 \le \frac{5}{2}\,\mathrm{Vol}_4(X), \qquad \rho(X) \le \frac{5}{4}.$$

Proof. First suppose X is smooth and uniformly convex. The quaternionic construction of Chaidez–Tanny associates a Finsler metric F on $\mathbb{RP}^2$ whose unit codisks are affine squared bodies. Their four-sheet contact covering gives

$$c_{\mathrm{EHZ}}(X) \le 4\ell(F), \qquad \mathrm{Vol}_4(X) = 4\mathcal{V}(F)$$

([4, Theorem 4.4 and equations (5.2)–(5.3)]). Hence

$$c_{\mathrm{EHZ}}(X)^2 \le 16\ell(F)^2 \le 10\mathcal{V}(F) = \frac{5}{2}\,\mathrm{Vol}_4(X).$$

For arbitrary convex X, use smooth uniformly convex approximants with the same finite symmetry. One obtains them by smoothing the invariant support function and adding a small Euclidean ball. Hausdorff continuity of volume and of EHZ capacity passes the bound to X, as in [4, Remark 5.4]. □

The planar constant five is sharp; neither corollary asserts a sharp constant for its geometric class. In particular the comparison between a geodesic and its closed lift, and the compatibility of codisks coming from one ambient body, may prevent simultaneous equality in the auxiliary estimates. The corollary gives the explicit constant $\sqrt{5}/2$ in a setting where [4, Theorem 11] supplies an unspecified constant below $\sqrt{3/2}$; it does not establish the Viterbo bound $\rho \le 1$ for I-invariant bodies.

9 Truncated parallelograms: strict Viterbo and competing covers

Balitskiy, Mitrofanov and Polyanskii conjectured a particular optimal trapezoidal partner for every square with one corner truncated [2, Section 6.2, Conjecture 6.1]. We construct a different trapezoidal partner which improves their proposed optimum on an open set of truncation parameters. This disproves the optimizer assertion of that conjecture. We first prove its strict-inequality assertion for every convex partner, so the two assertions have different outcomes.

We use their normal-triangle covering theorem [2, Theorem 3.3]: for a polygon Q, the inequality $c_{\mathrm{EHZ}}(K \times_L Q) \ge 1$ is equivalent to K containing a translate of each Q-normal triangle of unit Q-perimeter. Degenerate triangles, namely segments traversed in both directions, are included. The Q-perimeter of a closed polygonal curve with directed edge vectors e_i is $\sum_i h_Q(e_i)$, where h_Q denotes the support function. This quantity is unchanged by translating Q.

Choose

$$0 < A, B < 1, \qquad g = A + B - 1 > 0, \qquad a = 1 - A, \qquad d = B - a^2.$$

In particular $B > a$ and $d > 0$. Put $r = g/B$ and $s = g/A$, and define

$$Q_{A,B} = \mathrm{conv}\{(0,0), (1,0), (1,1-r), (1-s,1), (0,1)\}.$$

Both r, s belong to $(0,1)$, so this is a genuine corner-truncated square. Its area is

$$|Q_{A,B}| = 1 - \frac{rs}{2} = 1 - \frac{g^2}{2AB}. \tag{34}$$

The outward-normal directions of $Q_{A,B}$ are $\pm e_1, \pm e_2, (r,s)$. The only nondegenerate positively dependent triple is $-e_1, -e_2, (r,s)$; the only minimal positive pairs are the two opposite coordinate pairs. Consequently its complete list of unit normal templates, up to translation, consists of

$$T = \mathrm{conv}\{(0,0), (A,0), (A,B)\}, \quad -T, \quad H = [0, e_1], \quad V = [0, e_2]. \tag{35}$$

For completeness, traverse T in the order $(A,B), (A,0), (0,0)$. Its edge directions are

$$-e_2, \quad -e_1, \quad (r,s),$$

and its $Q_{A,B}$-perimeter is one: the first two support-function contributions vanish, while $h_{Q_{A,B}}(A,B) = 1$, because the cut line is $rx + sy = r + s - rs = g/(AB)$ and $(A,B) = (AB/g)(r,s)$. The two coordinate segments have unit perimeter because $Q_{A,B}$ has unit width in both coordinate directions. Reversing the order of the three edge directions gives the translation class $-T$.

9.1 The Viterbo inequality for all partners

The covering description also proves the Viterbo inequality for the entire corner-truncated-square class, independently of which cover has minimum area.

Theorem 9.1. *Let $Q \subset \mathbb{R}^2$ be a parallelogram with one corner cut off by a line meeting the interiors of its two adjacent sides. For every planar convex body K with nonempty interior,*

$$c_{\mathrm{EHZ}}(K \times_L Q)^2 < 2|K|\,|Q|.$$

In the normalization $Q = Q_{A,B}$ above, every capacity-one cover satisfies

$$|K| > \frac{AB}{2AB - (A+B-1)^2}. \tag{36}$$

Thus the strict-inequality assertion of [2, Conjecture 6.1] holds, although its proposed optimizer will be disproved below.

Proof. An affine change in the Q-plane, paired with its inverse transpose in the K-plane, reduces to $Q_{A,B}$ without changing the capacity or the product of areas. Scaling K, we may assume its product capacity is one. It therefore contains translated copies of the four templates $T, -T, H, V$ from (35).

First, this implies a lattice covering property. Put

$$\Lambda = A\mathbb{Z}e_1 + B\mathbb{Z}e_2.$$

Split every fundamental rectangle by its diagonal parallel to (A,B). The resulting triangulation has cells that are translates of T or $-T$. We use the triangulation-covering lemma [2,

Lemma 4.3]: if every cell of a planar triangulation fits by translation into a convex body, then every translate of that body meets the vertex set of the triangulation. It follows that

$$K + \Lambda = \mathbb{R}^2. \tag{37}$$

Suppose first that K contains a translate of the axis-parallel rectangle $[0, A] \times [0, B]$. Let w_x, w_y be the coordinate widths of K. The four caps based on the rectangle's sides and reaching support-maximizing points of K have pairwise disjoint interiors [2, Lemma 7.1]. Adding their areas to the rectangle's area gives

$$|K| \geq AB + \frac{B}{2}(w_x - A) + \frac{A}{2}(w_y - B) = \frac{Bw_x + Aw_y}{2} \geq \frac{A+B}{2},$$

because H, V fit into K and hence $w_x, w_y \geq 1$. The cap lemma applies to an arbitrary convex body: only the four support maximizers and the rectangle are used, so one may equivalently apply its polygonal statement to their convex hull. This branch is strictly stronger than the desired bound, since

$$\frac{A+B}{2} - \frac{AB}{2AB - g^2} = \frac{g(A(1-A) + B(1-B))}{2(2AB - g^2)} > 0.$$

Now suppose no translate of that rectangle is contained in K. For each $t \in \mathbb{R}^2$, form a finite graph whose vertices are $(K - t) \cap \Lambda$, joining two vertices when their difference is Ae_1 or Be_2. It is nonempty by (37). It is also a forest. Indeed, a cycle contains a simple rectilinear cycle; its bounded interior contains a fundamental lattice cell. The closure of that cell lies in the convex hull of the cycle vertices, hence in $K - t$. This would put a translate of $[0, A] \times [0, B]$ inside K, contrary to the present assumption. Thus, writing $N(t)$ for the number of vertices and $E_x(t), E_y(t)$ for the two types of edges,

$$E_x(t) + E_y(t) \leq N(t) - 1.$$

Integrate this inequality over a fundamental rectangle of Λ. Periodization of indicator functions gives

$$\int N(t)\, dt = |K|, \qquad \int E_x(t)\, dt = |K \cap (K + Ae_1)|, \qquad \int E_y(t)\, dt = |K \cap (K + Be_2)|.$$

The fundamental rectangle has area AB, so

$$|K| - AB \geq |K \cap (K + Ae_1)| + |K \cap (K + Be_2)|. \tag{38}$$

Finally, let $[u, u + e_1] \subset K$ be a placed copy of H. Convexity gives the homothetic inclusion

$$(1 - A)K + A(u + e_1) \subset K \cap (K + Ae_1),$$

because subtracting Ae_1 from the left-hand side gives $(1 - A)K + Au \subset K$. Consequently

$$|K \cap (K + Ae_1)| \geq (1 - A)^2 |K|.$$

The placed copy of V gives the analogous bound $|K \cap (K + Be_2)| \geq (1 - B)^2 |K|$. Inserting these inequalities into (38) yields

$$(1 - (1 - A)^2 - (1 - B)^2)|K| \geq AB.$$

The coefficient on the left equals $2AB - g^2 > 0$, which proves the weak form of (36). Together with $|Q_{A,B}| = (2AB - g^2)/(2AB)$, this proves the weak capacity-volume inequality for every partner.

It remains to exclude equality. Suppose $|K| = AB/(2AB - g^2)$. Replace K by the convex hull P of one placed copy of each of the four templates. This polygon has at most ten vertices, is contained in K, and still has product capacity at least one. The weak inequality already proved gives $|P| \geq AB/(2AB - g^2) = |K|$, hence $P = K$. Thus an equality body is a polygon. The rectangle case was strict, so equality must hold in the two homothetic overlap inclusions. In particular, with $[u, v] = [u, u + e_1] \subset K$,

$$K \cap (K + Ae_1) = (1 - A)K + Av. \tag{39}$$

The equality of sets follows from their inclusion and equal areas.

Let n be an outward normal of any genuine facet of K. The right side of (39) has a genuine facet with that normal. An edge of an intersection of polygons comes from an edge of one of the intersected polygons. Its supporting value is therefore $h_K(n) + A \min\{0, n \cdot e_1\}$. Comparing supporting values in (39) gives

$$(1 - A)h_K(n) + A\, n \cdot v = h_K(n) + A \min\{0, n \cdot e_1\},$$

so

$$h_K(n) = \max\{n \cdot u, n \cdot v\}.$$

Every facet of K consequently contains at least one endpoint of the unit horizontal segment. Each endpoint can belong to at most two genuine facets of a planar polygon. Hence K has at most four facets.

If K has four facets, their coverage by the two endpoints forces u, v to be opposite vertices. Applying the same argument to the vertical overlap equality shows that the endpoints of the unit vertical segment are the other two opposite vertices. The two diagonals are perpendicular and both have length one, so $|K| = 1/2$. This is impossible because $AB/(2AB - g^2) > 1/2$.

The only remaining case is a triangular K. After translation, write

$$K = M \operatorname{conv}\{0, e_1, e_2\}, \qquad \det M \neq 0.$$

The product symmetry $c(K \times_L Q) = c(Q \times_L K)$, followed by the inverse-transpose linear change in the other factor, shows that $M^T Q$ is a capacity-one partner of the standard triangle. Its two normal triangles triangulate the unit square grid. Applying the same triangulation-covering lemma therefore gives

$$Q + M^{-T}\mathbb{Z}^2 = \mathbb{R}^2.$$

This lattice has covolume

$$|\det M|^{-1} = \frac{1}{2|K|} = |Q|.$$

A lattice cover at this exact area has multiplicity one almost everywhere: its multiplicity is at least one and its integral over a fundamental cell is the cell's area. In particular distinct translates have disjoint interiors, so these translates tile the plane.

Every edge of a convex polygon that tiles by lattice translations has another edge parallel to it with the opposite outward normal. To see this, choose a relative-interior point of the edge which is not a vertex of any of the locally finitely many neighboring tiles. A tile covering points just beyond that edge must meet it in an edge with the opposite normal; interior overlap is impossible. But the truncation edge of $Q_{A,B}$ has no edge with the opposite normal. Thus $Q_{A,B}$ cannot be such a tile. This contradiction excludes equality and proves the strict inequality. □

The argument separates two possibilities: a full lattice cell already contained in K gives a direct cap-area estimate, while its absence makes the lattice-point graphs acyclic and converts lattice covering into an upper bound on overlap. The unit normal segments supply the opposite, homothetic lower bounds on those overlaps. Equality would force a partner with at most four sides; the four-sided case has too little area, and the triangular case would force the truncated square to tile by translations.

9.2 A uniform gap and nearby norms

The pointwise strict inequality has a useful uniform consequence. Here neighborhoods are taken in the Hausdorff topology on planar convex bodies; translations of either factor do not affect the capacity or the area.

Corollary 9.2. *Let Q be a genuine corner-truncated parallelogram. There are a constant $\delta_Q > 0$ and a neighborhood $\mathcal{U}_Q$ of Q such that, for every planar convex body K,*

$$c_{\mathrm{EHZ}}(K \times_L Q)^2 \le (1-\delta_Q)\, 2|K|\,|Q|,$$

and, for every $Q' \in \mathcal{U}_Q$,

$$c_{\mathrm{EHZ}}(K \times_L Q')^2 < 2|K|\,|Q'|.$$

In particular, this gives open sets of admissible factors containing pentagons without parallel sides and smooth strictly convex bodies.

Proof. Use the affine normalization and four normal templates of the preceding proof. Let m_Q be the infimum of $|K|$ over convex bodies containing translates of all four templates. This infimum is finite, and the normal cover criterion identifies the constraint with $c_{\mathrm{EHZ}}(K \times_L Q) \ge 1$.

We first check attainment. If K contains a vertical unit segment, the triangles joining that segment to points of minimum and maximum first coordinate have disjoint interiors. Consequently $\mathrm{width}_{e_1} K \le 2|K|$. The horizontal unit segment similarly gives $\mathrm{width}_{e_2} K \le 2|K|$. A minimizing sequence with bounded area is therefore uniformly bounded after translation. Blaschke selection provides a Hausdorff limit. The four template containments persist in the limit, since their translation vectors are bounded and may be passed to subsequences. In particular, the limit contains the nondegenerate triangle T and is a convex body. Continuity of area proves that it attains m_Q. The strict theorem, applied to this minimizer, gives

$$m_Q > \frac{1}{2|Q|}.$$

Indeed its capacity is at least one. Rescaling the first factor to capacity one now gives the first assertion with $\delta_Q = 1 - (2m_Q|Q|)^{-1} > 0$.

Translate Q so that the origin lies in its interior and choose $\varepsilon \in (0,1)$ small enough that

$$(1-\delta_Q)\left(\frac{1+\varepsilon}{1-\varepsilon}\right)^2 < 1.$$

For all Q' in a sufficiently small Hausdorff neighborhood of Q,

$$(1-\varepsilon)Q \subset Q' \subset (1+\varepsilon)Q.$$

Monotonicity and homogeneity of capacity, together with area monotonicity, yield

$$\frac{c_{\mathrm{EHZ}}(K \times_L Q')^2}{2|K|\,|Q'|} \le \left(\frac{1+\varepsilon}{1-\varepsilon}\right)^2 \frac{c_{\mathrm{EHZ}}(K \times_L Q)^2}{2|K|\,|Q|} < 1.$$

Small generic perturbations of the five vertices remove the parallel edge pairs, while smooth strictly convex bodies approximate Q in Hausdorff distance. Both therefore occur in this neighborhood. □

Remark 9.3. This argument establishes existence of the gap and of the neighborhood. It gives no explicit value of δ_Q or radius of $\mathcal{U}_Q$, and it does not identify the minimizing partner.

Theorem 9.4. *With the preceding notation, let*

$$D = \left(\frac{A^2 a}{d}, -\frac{A^2 B}{d}\right), \qquad L = -\frac{a}{A}D, \qquad R = L + (1,0), \qquad C = (A,B),$$

and let $K_{A,B} = \operatorname{conv}\{L, D, R, C\}$. *Then*

$$c_{\mathrm{EHZ}}(K_{A,B} \times_L Q_{A,B}) = 1, \tag{40}$$

$$|K_{A,B}| = F(A,B) := \frac{B(2A+B-1)}{2[B-(1-A)^2]}, \tag{41}$$

$$\rho(K_{A,B} \times_L Q_{A,B}) = \frac{2Ad}{(A+g)(2AB-g^2)} < 1. \tag{42}$$

The sides LD *and* RC *are parallel. Thus* $K_{A,B}$ *is a trapezoid.*

Proof. Consider the affine-linear functions

$$\phi(x,y) = Bx + ay, \qquad U(x,y) = y - \frac{g}{A}x - a, \qquad G(x,y) = y - \frac{AB}{g}(x-A).$$

Direct substitution, using $AB + ag = d$ and $d - Aa = g$, shows that

$$K_{A,B} = \{(x,y) : 0 \le \phi(x,y) \le B,\ U(x,y) \le 0,\ G(x,y) \ge 0\}. \tag{43}$$

Its successive vertices are L, D, R, C: LD and RC lie on the parallel lines $\phi = 0$ and $\phi = B$, respectively; LC lies on $U = 0$ and DR on $G = 0$. All four sides have positive length under the stated strict inequalities.

We exhibit placements of the four templates in (35). The triangle T is already contained in $K_{A,B}$. Indeed the values of (ϕ, U, G) at its successive vertices $(0,0), (A,0), (A,B)$ are

$$(0, -a, A^2B/g), \qquad (AB, -B, 0), \qquad (B, 0, B),$$

which satisfy (43). The horizontal segment is $L + H = [L, R]$. The vertical segment is $D + V$: at D its three values are $(0, -1, 0)$, and at $D + e_2$ they are $(a, 0, 1)$. Here $0 < a < B$.

For the opposite triangle, put

$$t = \left(A + \frac{aAg}{d}, \frac{aB(A+g)}{d}\right).$$

The values of (ϕ, U, G) at the vertices $t, t-(A,0), t-(A,B)$ of $-T + t$ are, respectively,

	ϕ	U	G
t	B	$-g$	aB
$t-(A,0)$	aB	0	$aB + A^2B/g$
$t-(A,B)$	0	$-B$	$AB(1-B)/g$

and again satisfy all four half-plane inequalities. The normal-triangle criterion therefore gives $c_{\mathrm{EHZ}}(K_{A,B} \times_L Q_{A,B}) \ge 1$.

The horizontal width of $K_{A,B}$ is exactly one. More explicitly,

$$L_x < D_x < A < R_x = L_x + 1,$$

since $A - D_x = Ag/d$ and $R_x - A = ag/d$ are positive. A horizontal segment longer than one cannot fit into $K_{A,B}$. Applying the same covering criterion after scaling thus gives the reverse capacity inequality, proving (40).

The horizontal diagonal LR has length one and splits the trapezoid into triangles with heights AB/d and Bg/d. Hence

$$|K_{A,B}| = \frac{AB + Bg}{2d} = \frac{B(A+g)}{2d},$$

which is (41). Combining this with (34) proves the displayed formula for the systolic ratio. Finally,

$$(A+g)(2AB - g^2) - 2Ad = (1-B)g^2 > 0,$$

so the ratio is strictly below one. □

9.3 Comparison with the proposed optimizer

The parameters used in [2, Section 6.2] are

$$\alpha = 1 - A, \qquad \beta = 1 - B, \qquad \gamma = g.$$

Indeed the cut line meets the coordinate axes at $(1/A, 0)$ and $(0, 1/B)$; the ratios defining α and β are therefore $1 - A$ and $1 - B$. Substituting these values into the proposed optimizer's systolic-ratio formula, and using (34), gives its capacity-one area

$$M(A,B) = \frac{AB(1-g^2)}{2(AB-g^2)}. \tag{44}$$

The denominator is positive throughout the parameter domain: for example, $g < A$ and $g < B$, so $g^2 < AB$.

The comparison factors exactly as

$$M(A,B) - F(A,B) = \frac{B(1-A)g^2(g-A^2)}{2d(AB-g^2)}. \tag{45}$$

Consequently the new trapezoid has strictly smaller capacity-one area, and strictly larger systolic ratio, whenever

$$B > 1 - A + A^2.$$

This is a nonempty open subset of the admissible parameters. Exchanging the coordinate axes and exchanging the two opposite triangle classes gives the symmetric competitor $F(B,A)$, which improves the proposed value whenever $A > 1 - B + B^2$.

For a concrete rational instance take $A = 11/20$ and $B = 4/5$. Then

$$Q = \operatorname{conv}\{(0,0), (1,0), (1,9/16), (4/11,1), (0,1)\}, \qquad |Q| = \frac{303}{352}.$$

The alternative trapezoid has vertices

$$L = \left(-\frac{891}{4780}, \frac{396}{1195}\right), \quad D = \left(\frac{1089}{4780}, -\frac{484}{1195}\right), \quad R = \left(\frac{3889}{4780}, \frac{396}{1195}\right), \quad C = \left(\frac{11}{20}, \frac{4}{5}\right).$$

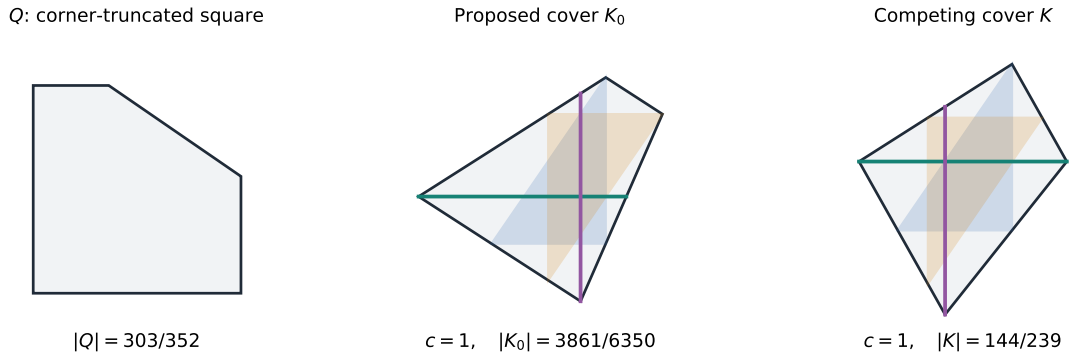


Figure 4: The corner-truncated square, the proposed partner, and the competing partner at the exact parameters $A = 11/20$, $B = 4/5$. The two cover panels use the same scale; each contains translations of the four unit normal templates.

The two capacity-one areas and systolic ratios are

$$|K| = \frac{144}{239} < \frac{3861}{6350} = |K_0|, \qquad \rho(K \times_L Q) = \frac{2629}{2727} > \frac{101600}{106353} = \rho(K_0 \times_L Q).$$

The ratio difference is exactly $931/106353$.

The theorem provides explicit competitors, not a classification of optimal partners. In particular it does not prove that $\min\{M(A,B), F(A,B), F(B,A)\}$ is the minimum possible area of a $Q_{A,B}$-cover. It also gives no universal product systolic bound. The remaining global covering question includes covers with arbitrarily many sides, although each cover can be replaced by the hull of placements of the ten vertices in (35).

9.4 An affine-area deformation through pentagonal covers

The comparison above comes from a deformation of the proposed cover itself. It is not merely a comparison between unrelated polygons. Write $e = AB - g^2 > 0$ and put

$$z_F = \frac{A^2 a}{d}, \qquad z_M = \frac{A^2 a}{e}.$$

For $z_F \le z \le z_M$, define

$$D_z = \left(z, \frac{g}{A} z - A\right), \quad L_z = -\frac{a}{A} D_z, \quad R_z = L_z + (1,0), \quad t_z = C + \frac{g}{A} D_z,$$

and let

$$P_z = \operatorname{conv}\{L_z, D_z, R_z, t_z, C\}, \qquad C = (A,B).$$

At the left endpoint R_z, t_z, C are collinear and $P_{z_F} = K_{A,B}$. At the right endpoint D_z, R_z, t_z are collinear and P_{z_M} is the proposed trapezoid K_0 of [2, Figure 11]. Indeed its ten template vertices obey exactly the three four-point collinearities used there to specify K_0: the line $L_z D_z$ contains 0 and $t_z - C$; the line $D_z t_z$ contains $(A,0)$ and R_z; and the line $L_z C$ contains $t_z - (A,0)$ and $D_z + e_2$.

Proposition 9.5. *For every $z \in [z_F, z_M]$,*

$$c_{\mathrm{EHZ}}(P_z \times_L Q_{A,B}) = 1, \qquad |P_z| = \frac{1 + Ag}{2} + \frac{g(g - A^2)}{2A^2} z. \tag{46}$$

For $z_F < z < z_M$, the five displayed vertices are genuine vertices of P_z. All moving templates undergo translations parallel to the same vector $(1, g/A)$.

Proof. The identities

$$d - e = gB > 0, \qquad e - ag = B(1-B) > 0, \qquad e - Aa = g(1-B) > 0$$

show that

$$0 < z_F < z_M < \min\{A, A^2/g\}.$$

The cross products of consecutive edge vectors of the displayed cycle L_z, D_z, R_z, t_z, C are, in that order,

$$1 - \frac{gz}{A^2}, \quad a - \frac{ez}{A^2}, \quad g\left(-a + \frac{dz}{A^2}\right), \quad g\left(A + \frac{az}{A}\right), \quad A + \frac{az}{A}.$$

They are nonnegative, with only the stated endpoint degeneracies. The vertical levels satisfy $D_{z,y} < L_{z,y} = R_{z,y} < t_{z,y} < C_y$; the two chains from D_z to C are consequently monotone, and the cycle is the boundary of the claimed convex polygon.

The four templates have the placements T, $-T + t_z$, $H + L_z$, and $V + D_z$. Their containment follows from the following segment identities:

$$0 = L_z + a(D_z - L_z), \qquad (A, 0) = D_z + A(R_z - D_z),$$
$$t_z - C = L_z + B(D_z - L_z), \qquad D_z + e_2 = L_z + \frac{z}{A^2 + az}(C - L_z),$$
$$t_z - (A, 0) = L_z + \frac{Bz}{A^2 + az}(C - L_z).$$

Every coefficient lies strictly between zero and one. Thus the covering theorem gives capacity at least one.

The horizontal chord $L_z R_z$ has length one and is a longest horizontal chord. Below its height the chord length increases linearly from zero. Immediately above it, the derivative of the chord length is negative: putting $v = t_z - R_z$ and $w = C - L_z$, both v_y, w_y are positive and

$$\frac{v_x}{v_y} - \frac{w_x}{w_y} = \frac{\det(v, w)}{v_y w_y} < 0, \qquad \det(v, w) = -\frac{ag(A^2 + az)}{A^2}.$$

Horizontal chord length is a concave function of height in a convex planar body. Hence no horizontal segment longer than one fits, which proves the reverse capacity inequality. Finally, the shoelace formula for the five displayed vertices gives (46). □

If $g > A^2$, moving from z_M toward z_F strictly decreases area while preserving capacity. Thus the proposed optimizer is not even a local minimum of capacity-one cover area in this parameter region. If $g < A^2$, the opposite endpoint minimizes area within this particular family. At $g = A^2$, the entire interval consists of equal-area, capacity-one covers, including genuine pentagons between the two trapezoidal endpoints. These assertions concern the explicit deformation; they do not establish global minimality of either endpoint or of the intermediate pentagons.

10 Methods, scope, and companion material

The action-body construction turns affine capacity optimization into a support-function and Pfaffian problem. It yields the full-affine pentagon bound and both simplex bounds. In four dimensions the simplex interaction graph admits a direct proof; in six dimensions the global equality theorem also depends on an exact cdd/PPL vertex classification. These are class bounds, while the HKO intersection uses a complete exact capacity computation at one moderate parameter. The separate first-variation reduction uses labeled shortest-orbit measures: seven selected cuts give a seventeen-facet ascent near H, while adding two arbitrary characteristic-invariant affine hinges and retaining the original facet inequalities cannot give positive first variation. The latter is a scoped first-order statement, not a local-maximality theorem for all twelve-facet bodies. Neither calculation determines the unrestricted four-dimensional optimum or the maximum along the twenty-facet family.

The planar results use geometric rather than action-polytope reductions. Shadow deformations and confocal paraboloids give a sharp difference-area constant for squared bodies, but the Chaidez–Tanny capacity transfer does not establish a sharp four-dimensional constant. Translation covers and lattice overlaps give a strict inequality for every partner of a truncated parallelogram, even though its proposed minimum-area partner fails. The actual minimizing partner and bounds for intersections of three or more confocal paraboloids remain open here.

Source and data availability. This PDF is report revision `2026-10-29-r6` and is to travel with its same-revision source-and-evidence companion archive. That archive supplies the LaTeX source, figure sources, exact inputs and replay programs under `certificates/`, scoped reviews, and an SHA-256 file manifest. The finite certificate arrays and full enumeration outputs are not printed in this PDF. The replay commands and software requirements are in the respective certificate READMEs. A successful replay checks the stated finite predicates; the geometric implications are the written proofs here.

A First variation under simultaneous facet splittings

Fixed-facet variations do not describe the birth of several facets at once. Here the relevant first-order datum is a convex function on each old facet: it records the maximum depth of the competing cuts. We obtain its exact capacity response from labeled occupation measures of shortest polygonal characteristics. The resulting convex-order criterion is a criterion for *first-order stationarity*. It does not prove a Hausdorff neighborhood of local maximality for the pentagon counterexample.

Let $K = \{x \in \mathbb{R}^{2n} : a_i \cdot x \le 1,\ 1 \le i \le m\}$ be a bounded polytope containing the origin in its interior, with genuine facets F_i. Here $\mathbb{J}$ is characterized by $\omega_0(u,v) = \langle \mathbb{J}u, v\rangle$, with $\mathbb{J}^2 = -I$. Its matrix is the negative of the skew-form matrix J in Section 2.1; throughout this appendix put $\omega = \omega_0$. For a finite word of facet labels and weights, put

$$Q(\sigma,\beta) = \sum_{r<s} \beta_r \beta_s \omega(a_{\sigma_r}, a_{\sigma_s}), \qquad \sum_r \beta_r a_{\sigma_r} = 0, \quad \sum_r \beta_r = 1, \quad \beta_r \ge 0. \tag{47}$$

Haim–Kislev's formula [6] gives $Q_* := \max Q = 1/(2c_{\mathrm{EHZ}}(K))$ using words without repeated labels. Repeated words also satisfy $Q \le Q_*$: their unit-period polar loop has velocity $2\mathbb{J}a_{\sigma_r}$ for time β_r, action $2Q$, and dual energy $\frac{1}{4}\int h_K(-\mathbb{J}\dot z)^2 = 1$. Normalizing its action to one in the dual action principle gives $c_{\mathrm{EHZ}}(K) \le 1/(2Q)$. Translating the loop to zero mean changes neither quantity. For redundant rows the support values are at most one, so the same inequality holds.

We impose the *full-rank support condition*: after deleting zero weights from any finite maximizing word, the augmented constraint columns span $\mathbb{R}^{2n+1}$, or equivalently

$$\operatorname{rank}\big[(a_{\sigma_r}, 1)\big]_r = 2n+1.$$

This concerns the constraint matrix, not the Hessian of the quadratic program.

A maximizing word reconstructs a shortest boundary characteristic, as shown below. Parametrize it on $[0,1]$ with velocity $2T\mathbb{J}a_{\sigma_r}$ for time β_r, where $T = c_{\mathrm{EHZ}}(K)$. Its labeled occupation measure μ_γ is a probability on the disjoint union $X = \bigsqcup_i F_i$: time on a segment with label i is assigned to that copy of F_i, including at ridges. Let $\mathcal{M}$ be the weak closure of the measures of *all finite maximizing words*, allowing repetitions. The normalized cone-volume probability is

$$d\nu|_{F_i} = \frac{dS}{2n\ \mathrm{Vol}_{2n}(K)|a_i|}. \tag{48}$$

Theorem A.1 (Simultaneous-splitting derivative)**.** *Assume the full-rank support condition. For each facet choose a nonempty finite list $D_{ij} \in \mathbb{R}^{2n}$ and set*

$$K_t = \{x : (a_i + tD_{ij}) \cdot x \le 1 \text{ for every } i,j\}, \qquad f_i(x) = \max_j D_{ij} \cdot x \quad (x \in F_i).$$

For this fixed finite list, the right derivative exists and equals

$$\left.\frac{d}{dt}\right|_{0+} \log \frac{c_{\mathrm{EHZ}}(K_t)^n}{n!\,\mathrm{Vol}_{2n}(K_t)} = 2n\left(\int_X f\,d\nu - \max_{\mu\in\mathcal{M}} \int_X f\,d\mu\right). \tag{49}$$

In particular, the hypothesis holds for every Lagrangian product of two planar polygons having no antipodal facet normals, including the HKO product.

Proof. We give the sensitivity and placement arguments explicitly, since neither requires a smooth optimizing branch.

A local chart at every limiting optimizer. For the fixed augmented row list, the moving optimum $q(t)$ is the maximum over finitely many permutations and their feasible weights.

This remains true with redundant or repeated presentation rows: genuine facets give one inequality and the polar-loop bound above gives the other. Along any sequence $t \downarrow 0$, compactness of the weight simplex and finiteness of the words give maximizing weights $\beta_t \to \beta_0$ in one word. Capacity continuity makes β_0 a base maximizer. Choose $2n+1$ positive coordinates I forming an invertible constraint minor, and denote the remaining weights by z. With columns $B_r(t) = (a_{\sigma_r} + tD_r, 1)$, solve

$$\beta_I(t,z) = B_I(t)^{-1}(e - B_J(t)z), \qquad e = (0,\dots,0,1)^T.$$

The solved weights stay positive near $(0, z_0)$; the other coordinates merely satisfy $z \geq 0$. In particular $\beta(0, z_t)$ is base-feasible. Writing $F(t,z) = Q_t(\beta(t,z))$, we obtain

$$q(t) - Q_* \leq F(t,z_t) - F(0,z_t) = t\,\partial_t F(0,z_0) + o(t).$$

Conversely the section with $z = z_0$ fixed gives the reverse lower bound for every base optimizer. Thus $q'(0+)$ exists and is the maximum of these section derivatives. This comparison also covers singular Hessians and continua of maximizing weights.

Placement of a maximizing word. Delete zero weights and write $a_r = a_{\sigma_r}$, $s_r = \sum_{k<r} \beta_k a_k$, and $p_r = \partial_{\beta_r} Q$. Stationarity and full rank give unique multipliers satisfying

$$p_r = \lambda \cdot a_r + \eta, \qquad \eta = 2Q_* = 1/T, \qquad p_r = -2\omega(a_r, s_r) = 2a_r \cdot \mathbb{J}s_r.$$

The vertices $x_r = -T\lambda + 2T\mathbb{J}s_r$ therefore satisfy $a_r \cdot x_r = 1$. The segment to x_{r+1} has direction $\mathbb{J}a_r$, so its active facet equation holds throughout. To obtain every other facet inequality, insert an arbitrary normal a_k at any point of the loop, subdividing its current segment if necessary. Give the inserted row weight $\varepsilon \geq 0$ and compensate with the positive minor. Its reduced objective derivative is

$$p_{\text{new}} - \lambda \cdot a_k - \eta = \frac{a_k \cdot x - 1}{T}.$$

Global optimality among repeated words makes this nonpositive. Hence the whole loop lies in K, on its labeled facets, and has action T.

The midpoint formula. Differentiate a feasible section of this word with row velocities D_r. The differentiated constraints give $\sum_r a_r \beta_r' = -\sum_r \beta_r D_r$ and $\sum_r \beta_r' = 0$. The direct row derivative of Q is

$$\sum_r \left[-2\beta_r \omega(D_r, s_r) - \beta_r^2 \omega(D_r, a_r)\right].$$

Adding the weight derivative $-\sum_r \beta_r \lambda \cdot D_r$ yields

$$\begin{aligned} Q' &= \frac{1}{T} \sum_r \beta_r D_r \cdot x_r^{\text{mid}}, \\ x_r^{\text{mid}} &= x_r + T\beta_r \mathbb{J}a_r. \end{aligned} \tag{50}$$

The self term in the preceding display is essential: the evaluation point is the segment midpoint, not its starting point. For every optimizer in the finite bank, $D_r \cdot x \leq f_i(x)$ on its segment labeled i. Affine midpoint integration therefore gives $q'(0+) \leq T^{-1} \max_{\mathcal{M}} \int f \, d\mu$. For the reverse inequality, take any finite maximizing base word and subdivide every segment at the finitely many switches of the maximum $f_i = \max_j D_{ij} \cdot x$. Assign a maximizing row on each resulting interval. After deleting zero-length pieces, this is a positive repeated word with the same base objective and full rank. A feasible continuation exists, and the polar-loop

inequality bounds its objective by $q(t)$. Equation (50) now integrates f exactly. Taking the supremum and then weak closure gives

$$q'(0+) = 2Q_* \max_{\mu\in\mathcal{M}} \int f\, d\mu, \qquad \frac{c'(0+)}{c(K)} = -2 \max_{\mu\in\mathcal{M}} \int f\, d\mu.$$

The maximum over $\mathcal{M}$ exists because X is compact.

Volume and the product hypothesis. Away from the ridges, the radial gauge of K_t has derivative f_i on F_i. The radial volume formula and dominated convergence, using the fixed finite row list, give

$$V'(0+) = -\sum_i \int_{F_i} \frac{f_i}{|a_i|}\, dS = -2nV \int f\, d\nu.$$

This proves (49). For a planar Lagrangian product, a positive-Q word must use both factors. Closure separates into the two planes. A positive closed combination of nonzero planar normals without antipodal pairs spans its plane; hence the active normals span $\mathbb{R}^4$. If $v \cdot a_r + s = 0$ on their positive support, weighting and summing gives $s = 0$, then $v = 0$. The augmented rank is therefore five, also for repeated words. □

Corollary A.2 (A common convex-order ensemble)**.** *Under the theorem's hypothesis, the following are equivalent:*

(i) *Every fixed finite simultaneous-splitting direction has nonpositive first systolic variation.*

(ii) *The weakly closed convex hull of $\mathcal{M}$ contains a probability $\bar{\mu}$ such that $\int f\, d\bar{\mu} \geq \int f\, d\nu$ for every continuous function convex on each facet copy.*

It is enough in (i) *to use inward cuts retaining every old facet. Any such $\bar{\mu}$ has the same mass and first moment as ν on each facet.*

Proof. Finite maxima of affine functions are uniformly dense in the continuous convex functions on a compact facet. Every affine function there is $D \cdot x$ because $a_i \cdot x = 1$. Thus the theorem turns (i) into $\sup_{\mathcal{M}} \int f\, d(\mu - \nu) \geq 0$ for every facetwise convex f. Let $B = \overline{\mathrm{conv}}\mathcal{M}$. For a finite list $f_1, \ldots, f_k$, the compact convex set of vectors $(\int f_j\, d(\mu - \nu))_{j=1}^k$, $\mu \in B$, meets the positive orthant. Otherwise separation supplies nonnegative coefficients α_j whose sum $\sum_j \alpha_j f_j$ contradicts the preceding inequality. Compactness and the finite-intersection property now give one common $\bar{\mu}$ for all f. The converse is immediate. Adding one common large constant to all facet profiles makes them nonnegative, allowing the zero row to be retained; the difference of two probability integrals annihilates that constant. Finally test both signs of facetwise affine functions to obtain the mass and first-moment identities. □

The common-ensemble requirement is stronger than finding a different successful orbit for each single cut. Nor does first-order nonincrease give an unrestricted neighborhood theorem: zero derivatives and estimates whose constants deteriorate with cut count or slope still require analysis.

A.1 Two limitations of finite orbit reductions

The HKO body is $H = P \times_L R(-\pi/2)P$, where P is the circumradius-one regular pentagon. Put $\alpha = \pi/5$, $\tau = \tan\alpha$, and

$$a_0 = (1, \tau, 0, 0), \qquad w = (-\sin\alpha, \cos\alpha).$$

On $F_0 = E_0 \times R(-\pi/2)P$, the functional $\ell(q, p) = w \cdot p$ attains its maximum one exactly on $E_0 \times \{w\}$.

Proposition A.3 (No finite pure-word ensemble is universal)**.** *No finite convex combination of occupation measures of finite pure facet-word characteristics of H dominates ν for every facetwise convex profile. Here a pure segment has velocity a positive multiple of one $\mathbb{J}a_i$ and carries that label; shortestness is not required.*

Proof. On a segment labeled zero, ℓ is constant since $w \cdot (1, \tau) = 0$. Its value is strictly below one: a nonconstant p-segment cannot lie in the singleton exposing slice at w. A finite bank therefore has a common bound $\ell \le h < 1$ on all its zero-labeled segments. For $h < r < 1$, the profile $f_0 = (\ell - r)_+$, $f_i = 0$ for $i \neq 0$, vanishes on the bank but has positive ν-integral. It is realized by the single extra row $a_0 + t(-r, -r\tau, w_1, w_2)$. More explicitly, for $\delta = 1 - r < 1 - \cos(2\alpha)$, the pentagon cap has slice width $2z \cot \alpha$, so

$$\int_X f \, d\nu = \frac{\cot \alpha}{30 \operatorname{area}(P)} \delta^3 > 0.$$

□

A separate computer-assisted theorem, sourced in Section 4, establishes that H is a Hausdorff local maximizer of ρ among convex four-polytopes with at most eleven genuine facets. The proposition shows why no fixed finite bank of pure facet-word measures can dominate all convex profiles. It is compatible with that eleven-facet theorem and with the twenty-facet ascent of Section 4. Infinite ensembles and weak limits remain available in Corollary A.2. For an action-T word, the wedge at w also gives the quantitative bound $1 - \ell \ge T\beta \sec \alpha \tan \alpha$ on a zero-labeled segment of weight β, by comparing its length with the wedge's slice width.

A second obstruction concerns eliminating repeated normals by moving their weights to an endpoint. That operation need not maximize a convex profile's occupation response, even along a family of genuine HKO minimizers. For the following explicit example set $r = \sqrt{5}$, $\tau = \sqrt{5 - 2r}$, $u = (3 - r)/2$, $v = \tau(1 + r)/2$, and $s = r - 1$. The ten normals are

$$(1, \tau, 0, 0),\ (-u, v, 0, 0),\ (-s, 0, 0, 0),\ (-u, -v, 0, 0),\ (1, -\tau, 0, 0),$$
$$(0, 0, \tau, -1),\ (0, 0, v, u),\ (0, 0, 0, s),\ (0, 0, -v, u),\ (0, 0, -\tau, -1).$$

Proposition A.4 (An interior response maximum)**.** *Let $b = (5 - r)/20$ and consider*

$$\sigma = (0, 6, 7, 3, 2, 8, 3, 9), \qquad 0 \le q \le 1,$$
$$\beta(q) = \left(\frac{r}{10}, \frac{5 + r}{40}, \frac{5 - r}{40}, b(1 - q), b, \frac{5 - r}{40}, bq, \frac{5 + r}{40} \right).$$

Every member is a maximizing word. For the facetwise convex profile $f_3(x) = |x - c_3|^2$, $f_i = 0$ otherwise, where $c_3 = (-1/4, \tau(\tau^2 - 9)/8, 0, 0)$ is the facet centroid, its response is

$$g(q) = \int f \, d\mu_q = \frac{2 - r}{4} q^2 + \frac{-19 + 9r}{16} q + \frac{195 - 77r}{240}.$$

Its maximum occurs at $q_* = (7 - r)/8$ and satisfies $g(q_*) - \max\{g(0), g(1)\} = (r - 1)/128 > 0$.

Proof. Substitution gives closure and $Q = \tau/5$ at both endpoints. The objective is affine in the transfer between two identical normals, since $\omega(a_3, a_3) = 0$, so it is constant throughout. This is the HKO optimum by [7]; the placement argument above applies. For a direct calculation of the response, the two row-three segment midpoints are

$$m_1(q) = \frac{1}{4}(-1 - r, -\tau(1 + r), \tau(r + q), 1 + rq),$$
$$m_2(q) = \frac{1}{4}(-1, -\tau(2 + r), \tau(q - 3 - r), 1 - r + rq).$$

Their weights are $b(1-q)$ and bq. Exact integration of a quadratic along a straight segment of weight β gives $\beta(|m-c_3|^2+T^2\beta^2|a_3|^2/3)$. Here $T=5/(2\tau)$ and $T^2b^2|a_3|^2/3=(5-r)/24$. Inserting the displayed midpoints yields g; differentiating this strictly concave quadratic proves the claimed maximum and gap. □

Uniform approximation of this profile by finite maxima of affine functions preserves the strict endpoint gap. Thus the phenomenon also occurs for a finite inward split bank. It refutes the proposed convexity argument for deleting repetitions; it does not show that the two displayed endpoint words are best among *all* simple HKO words, and does not exhibit an improving systolic variation.

B A quantitative near-zero remainder for the HKO cuts

The derivative argument can be made effective without enumerating all twenty-row permutations. Its numerical step is deliberately tiny and illustrates quantitative control of the true capacity. The separate exact computation in Theorem 4.4 gives a substantially larger ratio increase at a moderate parameter.

Proposition B.1 (Effective improvement)**.** *For the original family* (16)*, with all ten unit-speed cuts,*

$$\frac{\mathrm{sys}(H_t)}{\mathrm{sys}(H)} > 1+\frac{t}{200}, \qquad 0<t\le 10^{-30}.$$

We give the analytic estimates, including the uniform pivot and the passage from near-maximizers to exact base maximizers. The finite primitive action gap and rational constant comparisons have an exact replay; see `certificates/hko-ascent/README.md`.

Put $q(t)=1/(2c_{\mathrm{EHZ}}(H_t))$, $q_0=\tau/5$, and $V_0=\mathrm{Vol}_4(H)$. The twenty presentation rows have the form a_r+tD_r: one copy of each base normal has $D_r=0$, and its other copy has the velocity in (15). A simple word in this bank therefore contains at most two occurrences of each base normal. For an ordered word, let

$$B_r(t)=(a_r+tD_r,1), \qquad e=(0,0,0,0,1)^{\mathsf{T}}, \qquad Q_t(\beta)=\tfrac{1}{2}\beta^{\mathsf{T}}G_t\beta,$$

where G_t is the symmetric ordered-action matrix. Write D for the matrix whose columns are $(D_r,0)$. Feasibility means $B(t)\beta=e$ and $\beta\ge 0$. The Haim–Kislev formula supplies a maximizing simple word for every t, while $H_t\subset H$ gives $q(t)\ge q_0$. The all-word response bound proved in Lemma 4.2 and Proposition 4.3 implies that every exactly maximizing base word with a feasible smooth continuation has action derivative at most

$$g_*=\frac{2q_0}{125}. \tag{51}$$

Indeed, the midpoint identity (50) makes that derivative $2q_0\sum_r \beta_r D_r\cdot x_r^{\mathrm{mid}}$, bounded by $2q_0$ times the total radial-hinge response. Repeated occurrences are covered by the proved reduction, not by an assumption that all maximizers are simple. The exact primitive calculation also gives

$$\Delta=\frac{-\tau+3\tau^3}{20}>\frac{1}{50} \tag{52}$$

as the minimum positive loss from q_0 among the 3000 triangle-pair cyclic six-orders. These finite predicates are checked in the companion; the following implications are analytic.

Lemma B.2 (A positive pivot for moving near-maximizers)**.** *Let $0 \le t \le 10^{-3}$, and let a feasible nonnegative simple word on the twenty-row bank have $Q_t(\beta) \ge q_0/2$. Five positive coordinates I have weights greater than $1/4500$, and their augmented base columns form an invertible matrix $B_I(0)$ satisfying*

$$|\det B_I(0)| > \tfrac{1}{2}, \qquad \|B_I(0)^{-1}\|_2 < 66, \qquad \|B_I(0)^{-1}\|_\infty < 200.$$

Proof. Let $N = \sqrt{5}-1 = |a_r|$. Both retained and moved rows obey $|D_r| \le \sqrt{1+(3N/5)^2} < 5/4$, while $N < 5/4$. Bilinearity and $\sum_{r<s} \beta_r\beta_s \le 1/2$ give

$$|Q_t(\beta) - Q_0(\beta)| \le \frac{25}{16}t + \frac{25}{32}t^2 < \frac{1}{60} < \frac{q_0}{6}.$$

Here $Q_0(\beta)$ uses the base normals but need not have base closure. Thus $Q_0(\beta) > q_0/3 > 1/30$. If s is the total q-factor mass, write $Q_0 = s(1-s)B(x,y)$ for normalized occurrence distributions in the two factors. Each cross pairing has absolute value at most $N^2 < 25/16$, so $|B(x,y)| \le 25/16$ and

$$s, 1-s \ge s(1-s) > \frac{8}{375}.$$

Perturbed closure gives the base residual directly, including the retained rows with zero velocity:

$$\left\|\sum_r \beta_r a_r\right\| = t\left\|\sum_r \beta_r D_r\right\| \le \tfrac{5}{4}t \le 2t.$$

After division by either factor mass, the mean residual of its normalized pentagon-direction distribution is at most $2t/(8/375) \le 3/32 < 1/8$. Passing from normals of length $N > 1$ to unit normals only decreases it.

Aggregate each normalized factor distribution by its five normal directions and order those masses decreasingly. Let r be the sum of the three smallest. The resultant of the two largest unit directions has norm at least $(1-r)\cos 72^\circ > (1-r)/4$; the other three contribute at most r in the opposite direction. A mean norm below $1/8$ therefore forces $r > 1/16$. Each of the three largest masses is at least $r/3 > 1/48$. Multiplying by the factor mass gives aggregate weight greater than $1/2250$ on each chosen direction. At most two row occurrences represent one direction, so one occurrence has weight greater than $1/4500$.

Choose two distinct q directions and three distinct p directions among these. Their five augmented columns are independent. The absolute determinant is the product of a two-normal planar determinant $N^2|\sin(k\pi/5)| > 1/2$ and a three-normal affine determinant, which has one of the exact values $(5\tau+\tau^3)/2$ or $5\tau-\tau^3$, both greater than one. Thus the five-column determinant exceeds $1/2$. Each augmented column has norm $\sqrt{1+N^2} = \sqrt{7-2\sqrt{5}} < 8/5$. Hadamard's inequality bounds each cofactor by $(8/5)^4 < 6.56$ and the adjugate's Frobenius norm by 32.8. Dividing by the determinant yields $\|B_I(0)^{-1}\|_2 < 66$ and hence $\|B_I(0)^{-1}\|_\infty < 200$. The argument is independent of word order and includes zero-velocity old rows. For completeness, perturbing these five columns by tD_I changes their determinant by at most $5(5/4)t(1.61)^4 < 1/8$ on the stated interval; the moving minor also stays invertible. □

Proof of Proposition B.1. Fix $0 < t \le 10^{-30}$ and a simple word attaining $q(t)$, with weights β_t. It satisfies the lemma because $q(t) \ge q_0$. Set $M = 200$, and keep the nonpivot weights $z = (\beta_t)_J$ fixed. Solve the five pivot weights for $0 \le s \le t$ by

$$\beta_I(s) = B_I(s)^{-1}(e - B_J(s)z), \qquad \beta_J(s) = z. \tag{53}$$

Let $\beta_0 = \beta(0)$. Since $B(0)\beta_t = e - tD\beta_t$,

$$\beta_{0,I} - (\beta_t)_I = tB_I(0)^{-1}D\beta_t.$$

The last row of D is zero. Using $|D_r| < 2$ gives an infinity-norm change at most $2Mt$. Hence every pivot of β_0 exceeds $1/4500 - 400 \cdot 10^{-30} > c := 10^{-4}$. The other weights are unchanged; β_0 is a base-feasible probability vector on the same word.

Here are uniform chart bounds, valid more generally for $0 \le s \le 10^{-8}$. The infinity norm of the five-column perturbation matrix is at most ten. The Neumann bound gives $\|B_I(s)^{-1}\|_\infty \le 2M$. The exact difference and differentiated constraint equations are

$$\begin{aligned} \beta_I(s) - \beta_{0,I} &= -sB_I(s)^{-1}D\beta_0, \\ \beta_I'(s) &= -B_I(s)^{-1}D\beta(s), \\ \beta_I''(s) &= -2B_I(s)^{-1}D_I\beta_I'(s). \end{aligned}$$

Thus $\|\beta(s)\|_1 \le 1 + 20Ms < 2$, $\|\beta'(s)\|_1 \le 40M$, and $\|\beta''(s)\|_1 \le 1600M^2$. For the last bound, each coordinate of β_I' has magnitude at most $8M$, each coordinate of β_I'' at most $320M^2$, and there are five. These estimates concern the algebraic chart; its weights need not stay positive throughout the larger auxiliary interval.

Put $F(s) = Q_s(\beta(s))$. Each moving normal has norm below two there, so entries of G_s, G_s', G_s'' are bounded in absolute value by $4, 8, 8$, respectively. Differentiation gives

$$\begin{aligned} |F''(s)| &\le 4\big[(40M)^2 + 2(1600M^2)\big] \\ &\quad + 2 \cdot 8 \cdot (40M) \cdot 2 + \tfrac{1}{2} \cdot 8 \cdot 2^2 \\ &= 19200M^2 + 1280M + 16 < 2 \cdot 10^5 M^2. \end{aligned}$$

With $R = 10^5 M^2 = 4 \cdot 10^9$, Taylor's theorem yields

$$q(t) = F(t) \le Q_0(\beta_0) + t\, g(\beta_0) + Rt^2, \tag{54}$$

where g is the derivative at zero of the fixed-pivot algebraic chart. For any base-feasible probability vector β on this word,

$$g(\beta) = \tfrac{1}{2}\beta^{\mathsf{T}} G_0'\beta + (G_0\beta) \cdot u(\beta), \qquad u_I(\beta) = -B_I(0)^{-1}D\beta, \quad u_J = 0.$$

The bounds $\|u(\beta)\|_1 \le 10M$, $\max|G_0| \le 4$, and $\max|G_0'| \le 8$ imply

$$|g(\beta)| \le 4 + 40M \le 44M, \qquad |g(\beta) - g(\beta')| \le 88M\|\beta - \beta'\|_1 \tag{55}$$

for base-feasible probability vectors β, β' on this word. The direct quadratic term contributes at most $8\|\beta - \beta'\|_1$; the other term contributes at most $80M\|\beta - \beta'\|_1$.

Let $d = q_0 - Q_0(\beta_0) \ge 0$. From $q(t) \ge q_0$, (54), and (55),

$$d \le 44Mt + Rt^2 < 10^{-20}, \qquad Q_0(\beta_0) > q_0/2. \tag{56}$$

We now round β_0 to an exact maximizing base word without changing its order or losing the positive pivot.

Let s be its q-factor mass and let u, v be its normalized factor weights. Each occurrence-wise planar closure polytope has at most forty vertices: each vertex uses one of five closing triangles and one of at most two occurrences of each of its three normals. The same count holds for a subword, which defines a face of the full occurrence polytope. Write convex decompositions $u = \sum_i \lambda_i u_i$ and $v = \sum_j \gamma_j v_j$, with at most forty vertices per factor. If

q_{ij} is the action of triangle pair (u_i, v_j) at balanced factor mass in this fixed order, put $S = \sum_{ij} \lambda_i \gamma_j q_{ij}$. Then

$$Q_0(\beta_0) = 4s(1-s)S, \qquad d = 4q_0(s-\tfrac{1}{2})^2 + 4s(1-s)(q_0 - S). \tag{57}$$

Every $q_{ij} \le q_0$, and every strict loss is at least Δ from (52). As $Q_0(\beta_0) > q_0/2$, we have $4s(1-s) > 1/2$. If ε is the total product mass of nonwinning pairs,

$$\varepsilon \le \frac{q_0 - S}{\Delta} \le \frac{2d}{\Delta}, \qquad 2|s - \tfrac{1}{2}| \le \sqrt{d/q_0}. \tag{58}$$

For $\varepsilon > 0$, delete from each marginal every vertex whose coefficient is less than $2\sqrt{\varepsilon}$ and renormalize. At most forty vertices are deleted per marginal, so the discarded mass is at most $80\sqrt{\varepsilon} < 1$ by (56). A surviving bad pair would have product mass at least 4ε, contradicting the definition of ε. Thus all surviving pairs win. If $\varepsilon = 0$, keep the original positive supports. Denote the resulting mixtures by u^*, v^* and put $\beta^* = (u^*/2, v^*/2)$ on the same ordered occurrences. It is base-feasible and $Q_0(\beta^*) = q_0$. Each marginal changes by at most $160\sqrt{\varepsilon}$ in ℓ^1, so (58) gives

$$\|\beta_0 - \beta^*\|_1 \le 2|s - \tfrac{1}{2}| + \tfrac{1}{2}\|u - u^*\|_1 + \tfrac{1}{2}\|v - v^*\|_1 \le \left(\frac{1}{\sqrt{q_0}} + 160\sqrt{\frac{2}{\Delta}}\right)\sqrt{d} \le 2000\sqrt{d}. \tag{59}$$

By (56), this is less than $2 \cdot 10^{-7} < c/2$. Every selected pivot weight of β^* is therefore positive. The same fixed-pivot chart gives a feasible smooth continuation through this exact base maximizer. Its derivative obeys (51), and (55)–(59) give

$$g(\beta_0) \le g_* + A\sqrt{d}, \qquad A = 88M \cdot 2000 = 35\,200\,000.$$

This applies the response bound only to an exact maximizing word; the near-maximizer is handled by the quantified rounding.

Substituting into (54) and maximizing $-d + At\sqrt{d}$ over $d \ge 0$ by completing the square gives a bound for the *true global* action:

$$q(t) \le q_0 + g_* t + (A^2/4 + R)t^2 \le q_0 + \frac{2q_0}{125}t + 10^{15}t^2. \tag{60}$$

Indeed $A^2/4 + R = 309\,764\,000\,000\,000 < 10^{15}$. No selected feasible branch is substituted for the global optimum.

For volume, on an old facet the gauge of H_t is at least $1 + tf_i$, because the old and moved rows both occur. Other inequalities only increase the gauge. The normalized cone-volume measure ν of (48) and the radial volume formula yield

$$\frac{\mathrm{Vol}_4(H_t)}{V_0} \le \int (1+tf)^{-4}\, d\nu \le 1 - 4\nu[f]t + \frac{8}{5}t^2.$$

Here $0 \le f_i \le 2/5$, and the second derivative of $(1+u)^{-4}$ is at most twenty for $u \ge 0$. Put $a = 2/125$, $v = 4\nu[f]$, and use (20):

$$\eta := v - 2a = \frac{20 + 64\sqrt{5}}{9375} > \frac{1}{100}.$$

As $q_0 > 1/10$, (60) gives

$$\left(\frac{q(t)}{q_0}\right)^2 \le (1 + at + 10^{16}t^2)^2 \le 1 + 2at + 3 \cdot 10^{16}t^2.$$

Multiplying this nonnegative bound by the volume bound and discarding only negative cross terms gives

$$\frac{q(t)^2 \operatorname{Vol}_4(H_t)}{q_0^2 V_0} \le 1 - \eta t + 4 \cdot 10^{16} t^2 < 1 - \frac{t}{200},$$

because $4 \cdot 10^{16} t < 1/200$ on the stated interval. Finally $\operatorname{sys}(H_t) = 1/(8q(t)^2 \operatorname{Vol}_4(H_t))$, so taking reciprocals proves the proposition. □

At $t = 10^{-30}$ this gives only the certified lower bound $5 \cdot 10^{-33}$ on relative gain. The reusable ingredient is the uniform quadratic remainder for the global optimal action; determining the maximum along the family remains a separate problem.

C Two characteristic-invariant cuts on the HKO product

We prove Proposition 4.7. All old HKO inequalities remain in the moving bank. The exact family predicates and replay instructions are at `certificates/hko-two-hinge/README.md`.

Quotient coordinates and volume. The facet-transitive symmetries used in Proposition 4.5 reduce the proof to the q-facet labeled zero. Put

$$r = \sqrt{5}, \quad N = r - 1, \quad e = J_2 n_0 / N, \quad m = \frac{1 - \cos(\pi/5)}{2}, \quad h = \frac{1 + \cos(\pi/5)}{2},$$

and on that facet use

$$\xi = \frac{e \cdot q}{\sin(\pi/5)}, \qquad \eta = \frac{e \cdot p - m}{h}.$$

The quotient ranges over $[-1, 1]^2$. Every affine function invariant along $\mathbb{J}a_0$ is $\alpha\xi + \beta\eta + \gamma$: its normal coordinate is constant on the facet, and invariance excludes precisely the remaining partner coordinate. Under facet-uniform measure, ξ is uniform on $[-1, 1]$ and independent of η. Put

$$k = r - 2, \qquad a = \frac{5 - r}{20}, \quad b = \frac{r}{10}, \qquad A = \frac{a}{2}, \quad B = \frac{5 + r}{40}, \quad C = \frac{5 + 2r}{10}.$$

Projection of the partner pentagon gives the probability density

$$p(y) = \begin{cases} b(y + 3), & -1 \le y \le k, \\ C(1 - y), & k \le y \le 1, \end{cases} \qquad \mu := \int y p(y)\, dy = -1 + \frac{2r}{5}. \tag{61}$$

Indeed its vertex projection levels are $-1, k, 1$, and the transverse slice width is affine between them. The exact projection, normalization and mean are also checked in the accompanying four-family certificate. Each facet has cone-volume mass $1/10$, so

$$\nu(f_0) = \frac{1}{10} \mathbb{E}(\alpha\xi + \beta\eta + \gamma)_+. \tag{62}$$

Four literal shortest-word families. The following families realize each edge coordinate $\xi = \pm 1$ and every listed value of $\eta = y$. The facet-zero occupation weight is

$$\beta_L(y) = A(y + 3) \quad (-1 \le y \le k), \qquad \beta_U(y) = B(1 - y) \quad (k \le y \le 1). \tag{63}$$

At the common endpoint, $\beta_L(k) = b$ whereas $\beta_U(k) = a$; the two families need not join with the same weight. Here are their complete ordered words and parameter intervals:

family	ξ	$y(z), \quad 0 \le z \le 1$	σ
L_+	1	$-1 + (k+1)z$	$(0,1,7,6,3,2,9)$
L_-	-1	$-1 + (k+1)z$	$(0,6,3,2,9,8,4)$
U_+	1	$k + (1-k)z$	$(0,7,3,9,4,5,1)$
U_-	-1	$k + (1-k)z$	$(0,4,6,5,2,1,8)$

In each listed order, the full weight vector is $\beta(z) = (1-z)\beta^- + z\beta^+$, with

	β^-	β^+
L_+	$(a,a,a,a,b,0,b)$	$(b,0,a,a,a,a,b)$
L_-	$(a,b,0,b,a,a,a)$	$(b,b,a,a,a,a,0)$
U_+	$(a,b,b,a,0,a,a)$	$(0,b,a,a,a,a,b)$
U_-	$(a,a,a,a,b,0,b)$	$(0,b,a,a,a,a,b)$

These are catalogue charts 115, 148, 72 and 19, with their parameters oriented to make y increase. Direct exact arithmetic checks closure, nonnegative weights and $Q = \tau/5$ at both endpoints and the midpoint. The KKT reconstruction checks all facet inequalities at all loop vertices, $\xi = \pm 1$, the displayed affine y coordinate, and (63). Endpoint checks extend to the whole family: the weights and reconstructed positions are affine, and the objective is quadratic. Interior weights are positive. Only the existence of these four shortest families is needed below; no numerical search result is a premise.

Lemma C.1 (Scalar hinge bound)**.** *For $g(y) = (uy + v)_+$, set*

$$I(g) = \int_{-1}^{1} g(y)p(y)\,dy, \qquad M(g) = \max\left\{ \sup_{[-1,k]} A(y+3)g(y),\ \sup_{[k,1]} B(1-y)g(y) \right\}.$$

Then $I(g) \le 5M(g)$.

Proof. Constants follow from $M(g) \ge bg$ and $5b = r/2 > 1$. Positive homogeneity reduces the other cases to decreasing hinges $(z-y)_+$ and increasing hinges $(y-z)_+$, for real thresholds z.

For a decreasing hinge with $z \le -1$ the claim is trivial. On $[-1,1]$, convexity in z bounds its integral by the endpoint secant:

$$I(z) \le \frac{z+1}{2}(1-\mu), \qquad M(z) \ge 2A(z+1).$$

The required comparison follows from $20A - (1-\mu) = (5-r)/10 > 0$. For $z \ge 1$ the hinge is affine on the support and $I(z) = z - \mu$. Put $z_0 = 2k+3 = 2r-1$ and $z_* = 1 + 4r/5$. When $1 \le z \le z_0$, the lower-family point $y = (z-3)/2$ gives $M(z) \ge A(z+3)^2/4$ and

$$\frac{5A}{4}(z+3)^2 - (z-\mu) = \frac{5A}{4}(z-z_*)^2 \ge 0. \tag{64}$$

For $z \ge z_0$, use $y = k$ and $M(z) \ge b(z-k)$. The difference $5b(z-k) - (z-\mu)$ is increasing, since $5b - 1 > 0$, and is nonnegative at z_0 by (64).

For an increasing hinge with $z \ge 1$ the claim is trivial. If $z \le -1$, then $I(z) = \mu - z$ and $M(z) \ge b(k-z)$. The difference of these bounds is smallest at $z = -1$, where $5b(k+1) - (\mu+1) = (25-9r)/10 > 0$. For $-1 \le z \le k$, integration of the lower density gives

$$I(z) = \mu - z + b\left((z+1)^2 + \frac{(z+1)^3}{6}\right). \tag{65}$$

On $[-1,0]$, use $5M(z) \geq 5b(k-z)$. The difference from (65) is concave, with second derivative $-b(z+3) < 0$; its endpoint values are

$$\frac{25-9r}{10} > 0 \quad \text{and} \quad \frac{210-91r}{60} > 0.$$

It is therefore positive throughout that interval. On $[0,k]$, the upper-family point $y = (1+z)/2$ lies in $[k,1]$ and gives $5M(z) \geq 5B(1-z)^2/4$. The difference from (65) is again concave: its second derivative is at most $(25-19r)/80 < 0$. Its endpoint values are

$$\frac{555-233r}{480} > 0 \quad \text{and} \quad \frac{7}{12}B(1-k)^2 > 0.$$

Finally, for $k \leq z < 1$, only the upper density contributes, and

$$I(z) = \frac{C(1-z)^3}{6}, \qquad M(z) \geq \frac{B(1-z)^2}{4}, \qquad \frac{I(z)}{M(z)} \leq \frac{2C}{3B}(1-z) \leq \frac{8}{3} < 5,$$

where $(C/B)(1-k) = 4$. This covers all increasing hinges. □

Proof of Proposition 4.7. Write the selected facet hinge as $f_0 = (\alpha\xi + \beta\eta + \gamma)_+$. Convexity in ξ gives the endpoint bound

$$\mathbb{E}_\xi f_0(\xi, y) \leq \frac{f_0(-1,y) + f_0(1,y)}{2}.$$

For either edge sign, Lemma C.1 and the four actual shortest families give

$$\mathbb{E}_\eta f_0(\pm 1, \eta) \leq 5M(f_0(\pm 1, \cdot)) \leq 5R(f_0).$$

Thus $\mathbb{E} f_0 \leq 5R(f_0)$; equation (62) proves $R(f_0) \geq 2\nu(f_0)$. Symmetry covers every facet.

For two hinges on distinct facet copies, the actual profile is $f+g$. Nonnegativity yields

$$R(f+g) \geq \max\{R(f), R(g)\} \geq 2\max\{\nu(f), \nu(g)\} \geq \nu(f) + \nu(g).$$

If both cuts concern the same facet, their actual profile is $h = \max(f,g)$. Then

$$R(h) \geq \max\{R(f), R(g)\} \geq 2\max\{\nu(f), \nu(g)\} \geq \nu(f) + \nu(g) \geq \nu(h).$$

The simultaneous-splitting derivative theorem gives nonpositive first variation in either case, with all old rows retained as stipulated. □

The responding shortest word may depend on the hinge. No common occupation ensemble is asserted. The factor two is what makes these separate responses sufficient to handle two simultaneous cuts.

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